REPORT

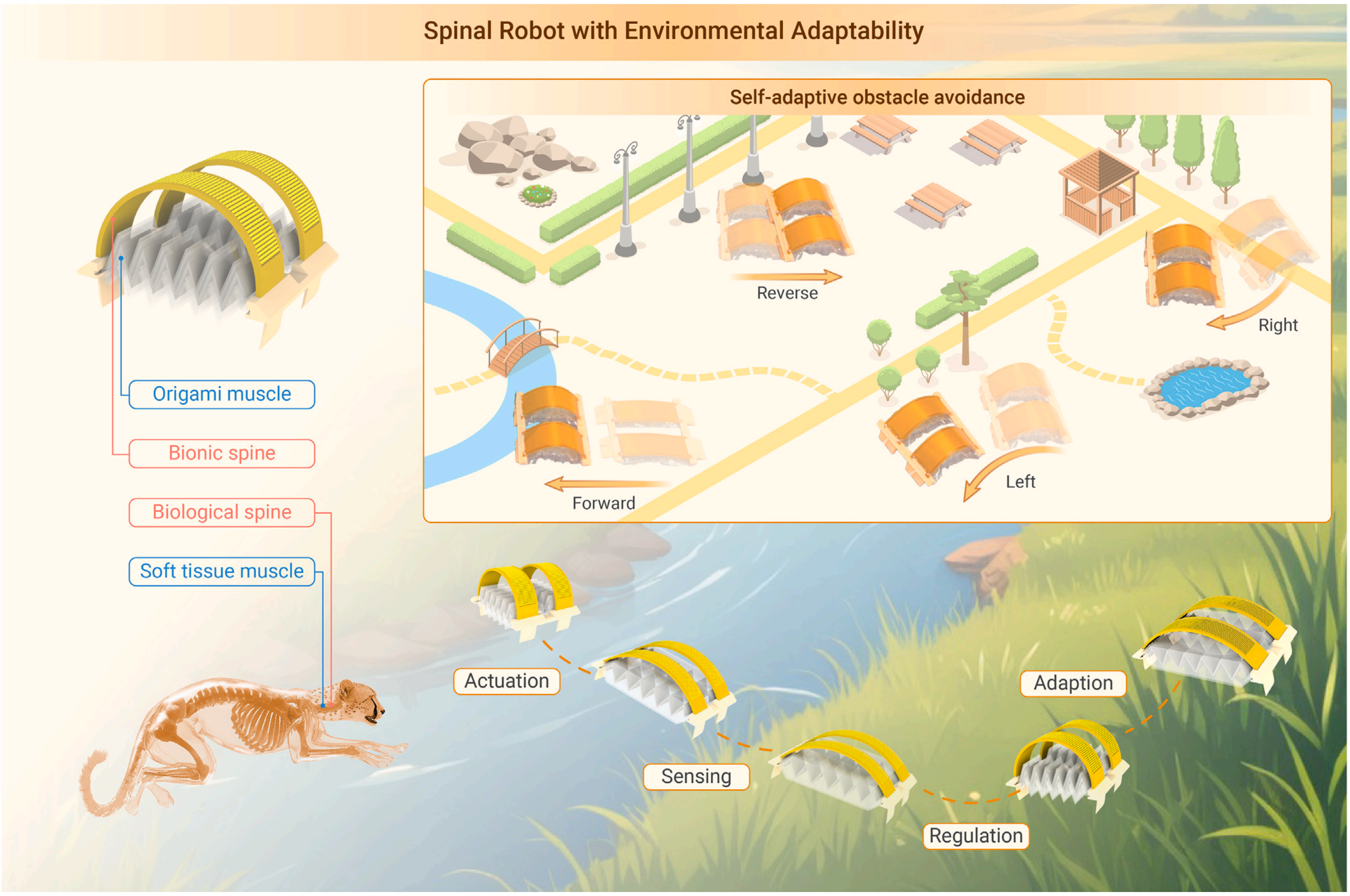
An intelligent spinal soft robot with self-sensing adaptability

Shoulu Gong,[1,2,4] Fuyi Fang,[1,4] Zhiran Yi,[1,*] Bohan Feng,[2] Anyu Li,[2] Wenbo Li,[3] Lei Shao,[2,*] and Wenming Zhang[1,*]

*Correspondence: yizhiran@sjtu.edu.cn (Z.Y.); lei.shao@sjtu.edu.cn (L.S.); wenmingz@sjtu.edu.cn (W.Z.)

Received: December 13, 2023; Accepted: May 15, 2024; Published Online: May 21, 2024; https://doi.org/10.1016/j.xinn.2024.100640

GRAPHICAL ABSTRACT



PUBLIC SUMMARY

- Intelligent spinal robots coupling sensing, recognition, and active adaption.
- A bionic spine with integrated sensing and actuation in one shared device.
- A design concept highly scalable for various robots for active environmental adaption and obstacle avoidance purposes.

An intelligent spinal soft robot with self-sensing adaptability

Shoulu Gong,[1,2,4] Fuyi Fang,[1,4] Zhiran Yi,[1,*] Bohan Feng,[2] Anyu Li,[2] Wenbo Li,[3] Lei Shao,[2,*] and Wenming Zhang[1,*]

[1]State Key Laboratory of Mechanical System and Vibration, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China

[2]University of Michigan-Shanghai Jiao Tong University Joint Institute, Shanghai Jiao Tong University, Shanghai 200240, China

[3]School of Aerospace Engineering and Applied Mechanics, Tongji University, Shanghai 200092, China

[4]These authors contributed equally

*Correspondence: yizhiran@sjtu.edu.cn (Z.Y.); lei.shao@sjtu.edu.cn (L.S.); wenmingz@sjtu.edu.cn (W.Z.)

Received: December 13, 2023; Accepted: May 15, 2024; Published Online: May 21, 2024; https://doi.org/10.1016/j.xinn.2024.100640

Citation: Gong S., Fang F., Yi Z., et al., (2024). An intelligent spinal soft robot with self-sensing adaptability. The Innovation **5(4)**, 100640.

Self-sensing adaptability is a high-level intelligence in living creatures and is highly desired for their biomimetic soft robots for efficient interaction with the surroundings. Self-sensing adaptability can be achieved in soft robots by the integration of sensors and actuators. However, current strategies simply assemble discrete sensors and actuators into one robotic system and, thus, dilute their synergistic and complementary connections, causing low-level adaptability and poor decision-making capability. Here, inspired by vertebrate animals supported by highly evolved backbones, we propose a concept of a bionic spine that integrates sensing and actuation into one shared body based on the reversible piezoelectric effect and a decoupling mechanism to extract the environmental feedback. We demonstrate that the soft robots equipped with the bionic spines feature locomotion speed improvements between 39.5% and 80% for various environmental terrains. More importantly, it can also enable the robots to accurately recognize and actively adapt to changing environments with obstacle avoidance capability by learning-based gait adjustments. We envision that the proposed bionic spine could serve as a building block for locomotive soft robots toward more intelligent machine-environment interactions in the future.

INTRODUCTION

Soft robots are promising for agile locomotion and high-level intelligence like living creatures and have been developed in a wide range of applications, including human-machine and machine-environment interactions.[1-6] It is worth noting that among living creatures, vertebrate animals not only feature highly developed muscular and skeletal systems but also an advanced central nervous system enclosed within the spines, which helps them to react very quickly to changes in their surroundings. This gives them a competitive edge compared to most invertebrates, which behave almost entirely by instinct and with little capability to learn from their mistakes. Such sophisticated musculoskeletal and nervous systems supported by the spines endow vertebrates the exceptional capacity to choose the optimal solutions and execute adaption when facing different survival environments.

There has recently been a great amount of research effort to develop intelligent adaption capabilities in biomimetic soft robots. Distinguished from the environmental adaption capabilities[7-11] that have been developed in current soft robots, such as simple phototaxis effect and non-perceptual adaption via shape morphing,[12,13] the environment adaption of vertebrates encompasses the entire process of perceptual feedback and autonomous adaption, which is a typical representation of high-level biological intelligence[14-16] by long-term evolution. Such adaption mainly relies on two functions of the biological spine: one the perception capability based on the enclosed central nervous systems and the other the regulatory adaption capability supported by the musculoskeletal systems. This high-level intelligence is the key to higher task success rate and efficiency. Therefore, to achieve such a level of intelligence in soft robots, it is highly desired to construct an entirely sensor-actuator-integrated spinal soft robotic system.

Current research effort about sensor-actuator-integrated soft robots mainly focuses on sensing based on e-skins[17-20] or visual systems[21-26] with applications in surrounding detection (e.g., pH, ion, oil, blow flow, etc.),[27-34] self-body perception (e.g., physical interaction monitoring, self-motion monitoring, etc.),[35-39] and intelligent recognition (e.g., environmental awareness, objects classification, etc.).[40-44] However, these existing sensor-actuator integration strategies only provide soft robotic systems with a perception capability but with unsatisfactory improvement on their motional performance and adaptability for various environments,[14,15,45] because their sensor and actuator components are discrete and independent devices with inadequate synergistic and complementary interaction.[46-48] Challenges remain in achieving an expected high-level intelligence where soft robots could be aware of the environmental terrains from attempted locomotion and then make self-adjustments autonomously, followed by learning and memorization.

This study proposes an intelligent soft robot with a sensing-actuation-integrated unibody spine for environmental recognition and active adaption. A flexible piezoelectric macro-fiber composite sheet is developed as a bionic spine, acting as the core component built upon an origami pneumatic robot. This bionic spine works as the nexus between environmental awareness and active adaption based on two reversible physical effects (shown in Figure 1A) that are successfully decoupled by a full bridge circuit. For one thing, the piezoelectric effect of the bionic spine provides reliable sensing for different environmental terrains as back-action; for another, the auxiliary actuation functions of the bionic spine enhance the actuation performance of the pneumatic artificial muscles by the converse piezoelectric effect, allowing the soft robot to travel more efficiently in various environments. Hence, the proposed soft robotic system enables both environmental recognition and autonomous adaption capabilities, showing higher task success rate and efficiency. In the following sections, we elaborate on the design of the spinal soft robots, characterize their motion enhancement, and then show the process of environmental recognition by machine learning. Finally, we demonstrate that the robot can perform various highly efficient and even beyond-biological locomotion with self-sensing adaptability, such as multi-terrain transitions, autonomous obstacle avoidance, and self-adaptive amphibious motions, which stand out from all previous sensor-actuator-integrated soft robots.

RESULTS

Design and methodology

As shown in Figure 1B, the robot structure contains an origami pneumatic actuator as the soft artificial muscle, a piezoelectric macro-fiber composite sheet as the bionic spine, and two printed resin robot feet. The origami actuator is vacuum driven with several advantages, including light weight, easy manufacturing, and large linear contraction rate (Figure S2 and Text S1). The bionic spine serves for both actuation and sensing purposes. It can be utilized as a supporting and auxiliary actuation based on converse piezoelectricity to enhance the motion performance over different environments, while it can also be utilized as a robotic e-skin simultaneously. The feedback signals of the bionic spine based on direct piezoelectric effect are available for analyzing both the motion states and the environmental interactions of the robot.

The self-sensing adaption process between the robot and the environment can be divided into three stages. Below, we take the amphibious transition of the robot between the land and the aquatic area as an example. During stage (1), the motion states of the robot and its interaction with the environment are constantly monitored by the bionic spine; in stage (2), changes in the monitored feedback signals are detected during terrain change and utilized for terrain recognition based on a pre-trained machine learning model; in stage (3), the actuation state of the robot will be self-regulated to actively adapt to the newly recognized terrain, thus switching to an optimal motion solution for this task. To achieve such an adaption, both enhanced actuation efficiency and accurate terrain recognition are necessary. Most importantly, a bridge circuit is developed to eliminate the crosstalk inside the bionic spine between the feedback signal as a sensor and the excitation signal as an auxiliary actuation, finally realizing the proposed unibody sensor-actuator integration design (more in Text S3).

Characterization of the spinal auxiliary actuation

As shown in Figure 2A, we first construct a soft crawling robot using a single pneumatic artificial muscle to demonstrate the design concept of the bionic

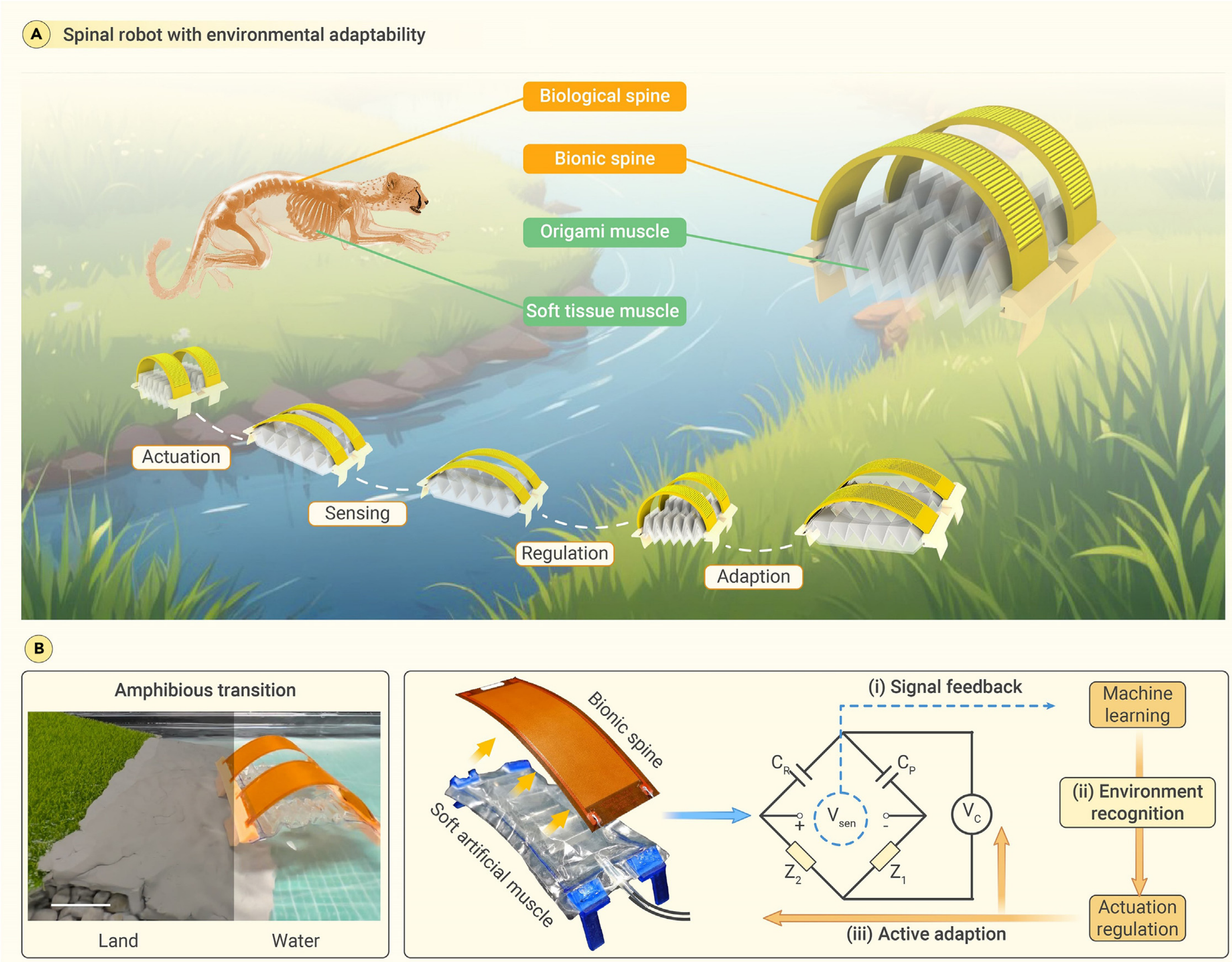


Figure 1. Depiction of high-level intelligence in both living creatures and soft robots (A) Spines for self-sensing, regulation, and environmental adaption. Vertebrate animals rely on spines for sensing and supporting and muscles for actuation. Proposed soft robots rely on bionic spines for sensing, support, and auxiliary actuation and origami muscles for actuation. (B) Demonstration of an autonomous self-sensing adaption process and methodology on the decoupling mechanism of a unibody sensing-actuation-integrated spine. Scale bar: 5 cm.

spine. The pneumatic actuation of the soft robot consists of two stages (more in Figure S6 and Text S1). First, when a negative pressure is applied, the artificial muscle contracts, and the deflection of the bionic spine increases, causing a simultaneous increase of the potential energy in both the bionic spine and the PET plastic skeleton inside the artificial muscle. As a result, the crawling robot will transit from the initial state to a contracted state. Second, when the negative pressure is withdrawn and a voltage is applied to the bionic spine to act as an auxiliary actuation, the crawling robot will revert to the initial state. The amplitude of the actuation pressure is set at −50 kPa to ensure dynamic stability of this soft crawling robot at different actuation frequencies. During the second stage, the potential energy of the PET plastic skeleton and bionic spine is released, leading to the generation of recovery forces $F_{Elastic}$ and $F_{Release}$ (more detail in Text S1), which can be expressed as follows:

$$F_{Elastic} = C_1 - C_2 x_a \qquad \text{(Equation 1)}$$

$$F_{Release} = K_1 \left[\int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1 - \sin^2\frac{\partial}{2}\sin^2\varphi}} \right]^2 \qquad \text{(Equation 2)}$$

As a result, the output force $F_{Auxiliary}$ generated by the bionic spine further facilitates the transition between the two motion states of the crawling robot, which can be expressed by the following equation:

$$F_{Auxiliary} = K_2 \frac{V}{y_a + \frac{x_a}{\tan\partial}} \qquad \text{(Equation 3)}$$

In the above equations, ∂ represents the central angle of the bent spine in a shape of a circularly arc, which significantly influences all output forces, C_1 and C_2 contain the intrinsic properties (sizes and Young's modulus) of the origami skeleton, K_1 and K_2 contain the properties (sizes, Young's modulus, and piezoelectric strain constant) of the bionic spine, x_a and y_a portray the shape of the curved bionic spine, and V is the applied voltage to the piezoelectric spine. Detailed derivations and modeling are available in Equations S1–S29.

The overall actuation force available in the second dynamic stage affects the crawling stride length of the robot. As an essential constituent of the overall actuation strategy, the auxiliary actuation is based on the converse piezoelectric effect of the bionic spine, and thus the amplitude of $F_{Auxiliary}$ is directly related to the magnitude of the excitation voltage. Figure 2B shows the normalized force amplitude of the soft crawling robot from the contracted state to the initial state (with a pneumatic actuation frequency of 1 Hz) as a function of the excitation voltage from 0.4 kV to 1 kV at the excitation frequencies of 1 Hz, 5 Hz, and

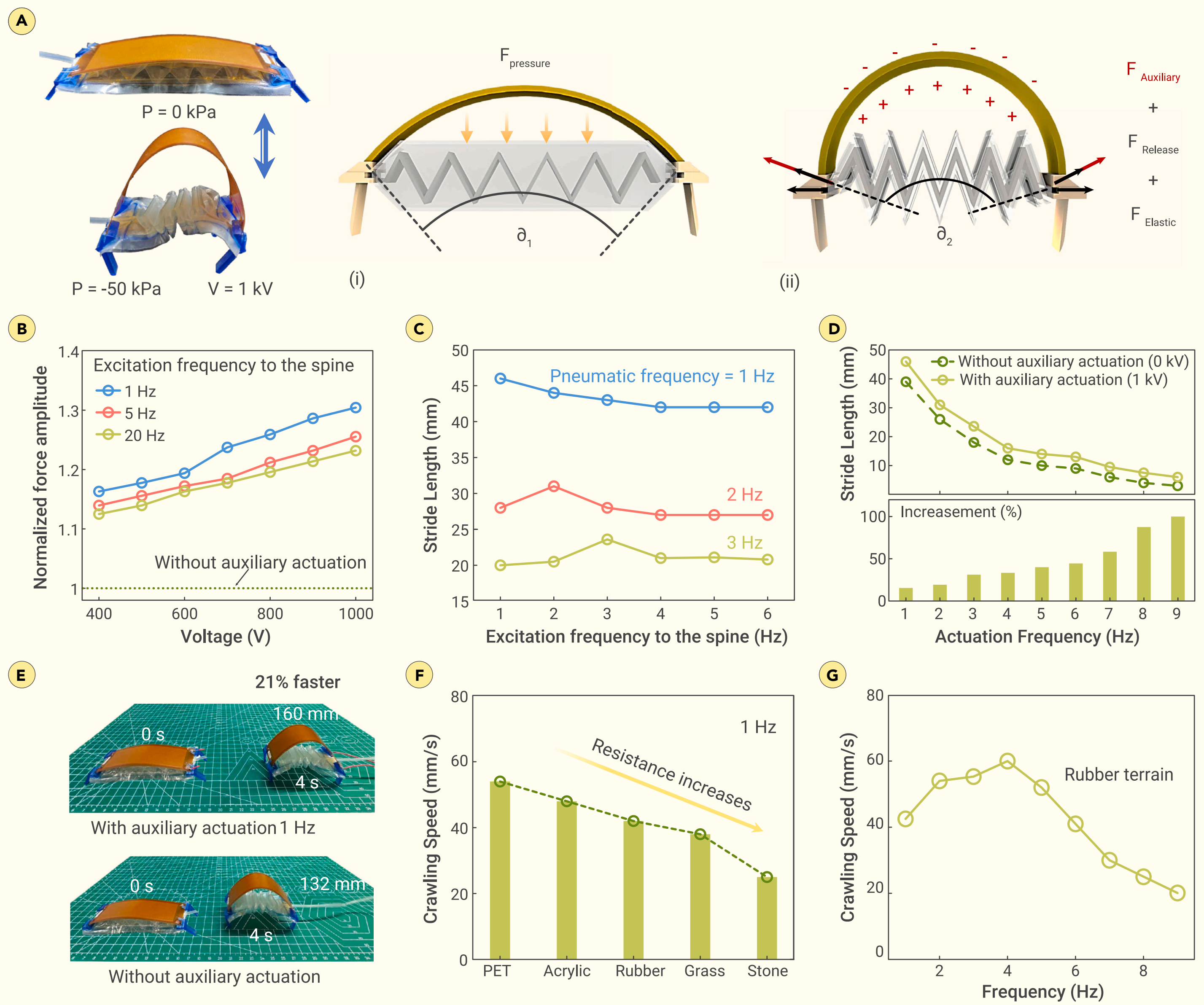


Figure 2. Actuation mechanism and characterization results of the spine-assisted actuation (A) Illustration of the actuation mechanism. (i) Contraction of the artificial muscle under vacuum pressure; (ii) recovering under the elastic force, release force, and auxiliary force generated by the artificial muscle and bionic spine. (B) Normalized recovery force under auxiliary actuation as a function of actuation voltage and actuation frequency, i.e., 1 Hz, 5 Hz, and 20 Hz from top to bottom, with the pneumatic actuation at 1 Hz. (C) Measured stride length of the crawling robot without terrain contact as a function of auxiliary frequency for three different pneumatic actuation frequencies, i.e., 1 Hz, 2 Hz, and 3 Hz. (D) Measured stride length of the robot as a function of actuation frequency with and without auxiliary actuation while the auxiliary voltage is 1 kV. (E) Comparison of robot crawling speed with and without auxiliary actuation on the rubber land. (F) Measured crawling speed of the robot at an actuation frequency of 1 Hz for different terrains. (G) Measured crawling speed of the robot as a function of actuation frequency on the rubber land.

20 Hz. The experimental results demonstrate that the presence of the auxiliary actuation greatly increases the overall output force in the second stage by approximately above 10%. The effectiveness of the auxiliary actuation increases linearly with voltage amplitude at a growth rate of approximately 1% per 0.1 kV, and the output force also performs with a similar trend at different excitation frequencies.

To achieve a synergistic mechanism between the auxiliary actuation of the bionic spine and the pneumatic actuation of the artificial muscle, we further optimize the actuation parameters. We first explore the association between different actuation parameters and the stride length of the crawling robot. Figure 2C shows the measured stride length of the crawling robot without terrain contacts as a function of the excitation voltage frequency for three different pneumatic actuation frequencies, 1 Hz, 2 Hz, and 3 Hz, indicating that the maximum stride length always occurs at the condition when the pneumatic and the auxiliary actuations are at the same frequency. In contrast to purely pneumatic actuation, the crawling robot with synergistic actuation outperforms by more than 20% in the stride length for the frequencies from 1 Hz to 9 Hz, as shown in Figure 2D. Moreover, the increments in stride lengths are especially pronounced (over 50%) when the actuation frequency exceeds 3 Hz, which is mainly because the auxiliary actuation force $F_{Auxiliary}$ greatly compensates the response hysteresis and low output force of the pneumatic artificial muscle at high frequencies. Figure 2E shows that the crawling robot with the auxiliary actuation has a 21% speed improvement from 33 mm/s to 40 mm/s moving on a green cutting board (a rubber-surfaced terrain). We note that although the auxiliary actuation could improve the locomotion speed, the actual crawling speed is still determined by both its dynamic states and the topographic characteristics of the contacting terrains.

Crawling speed of the robot on different terrains is a function of the actual stride length and actuation frequency. In general, both large motion stride length and high actuation frequency are beneficial for a faster locomotion but cannot be easily achieved simultaneously because the actuation frequency of the robot is inversely proportional to its stride length. Meanwhile, the resistance of the environmental terrain could reduce the stride length, causing a great variance of the crawling speed on different terrains. Figure 2F shows the measured crawling speed with a synergistic actuation frequency of 1 Hz on different terrains, PET plastic, acrylic, rubber, grass, and stone from left to right while with increasing resistance in order (roughness from 10 μm to 3,000 μm), indicating a significantly varied crawling speed. It is important to understand that the effect of the contact resistance on locomotion speed is not always negative. For instance, a properly large contact resistance can actually prevent slipping, which greatly improves the crawling speed of the robot under high-frequency actuation. Hence, we consider that for each specific terrain, there is always a corresponding set of optimal actuation parameters for a better efficiency and a higher speed, such as the actuation frequency mentioned in this section. As shown in Figure 2G, the robot demonstrates the highest crawling speed with an optimal actuation frequency of 4 Hz on the rubber land, which is about 39.5% faster than the speed under 1 Hz. With similar experimental steps, we also obtained the optimal actuation frequency of the crawling robot on other terrains, including grass land, stone land, gravel land, PET land, plush land, and acrylic land (Figure S7), and these optimal actuation frequencies bring significant speed improvement ranging from 39.5% to 80% for the aforementioned terrains (Table S1). We believe that the robot possesses the potential to always maintain efficient motion over different terrains if knowing the type of the terrains beforehand, which could be achieved by combining sensing and autonomous adjustment.

Environment sensing and recognition by the spine

Regarding highly desired perception, the bionic spine endows this soft crawling robot with a self-sensing capability for both proprioception and exteroception using the decoupled piezoelectric feedback signal by the bridge circuit. We first demonstrate the proprioception capability of the robot by monitoring the feedback signal from the bionic spine during motion cycles under different actuation frequencies. As shown in Figure 3A, the measured signal can intuitively provide feedback on the change in crawling frequency and stride length. The motion states of the crawling robot can be then reconstructed based on the proposed model relating the responsive signal and the spinal bending angles (Figure S8 and Text S2), thus providing the crawling robot with the awareness of the position and movement of its own body.

In the above discussion, we have tested the relationship between the actual stride length of the crawling robot and the terrain resistance (more in Figure S7). As shown in Figure 3B, for some relatively smooth terrains, the difference in resistance can be observed by small features of the spinal feedback signal. For other terrains with uneven surface structures, the topographic differences expressed in the spinal feedback signal are much more pronounced (more in Figure S10). Thus, the exteroception capability of the crawling robot for different terrains can be realized by analyzing features in the feedback signal, including plush, grass, stone, gravel, acrylic, and rubber lands, as shown in Figure 3C.

To distinguish the proprioception and the exteroception information in the same piezoelectric signal, machine learning techniques are utilized in processing the feedback signal for further robot self-learning and adaption. The whole modeling and prediction process is shown in Figure 3D (more detail in materials and methods). Figure 3E shows that seven terrains with different roughness and morphology can be recognized with a high accuracy of 98.5% by the crawling robot based on the bionic spine and machine learning techniques. To validate the reliability of the trained learning model, the crawling robot is set to crawl continuously across multiple terrains while different kinds of multi-terrain combinations are used (Figure S11). The crawling robot shows a high identification accuracy of 97.1% for multi-terrain tasks (Figure 3F).

Multi-terrain active adaption of the soft spinal robot

With the well-trained machine learning network, the intelligence of the soft spinal robot is further illustrated by demonstrating its self-sensing adaptability to different terrains. Adaption means that the robot can always perform with optimal motion efficiency over different terrains, which is significant for crawling robots in the wild. More important, it greatly improves the success rate of the crawling robot to cope with multi-terrain tasks, where it may fail to move over changing terrains without autonomous adaption. Below, we show the superiority of the self-adaptive crawling robot by comparing its performance with a unimodal one under multi-terrain tasks.

The first task contains three terrains, which are rubber, stone, and PET plastic lands representing elastic, rough, and slippery conditions in sequence. We record the whole process of the crawling robot performing the tasks while monitoring the readouts of its bionic spine (Figure S12B). As shown in Figure 4A, the unimodal crawling robot passes the rubber land smoothly from 0 s to 4 s (with a fixed actuation frequency of 4 Hz, which is the optimal motion frequency for the crawling robot on the rubber terrain), but it failed to deal with the transition from the rubber land to the stone land. This failure is mainly due to the low output force and small stride length of the unimodal crawling robot at the frequency of 4 Hz, which prevents the robot from crossing the obstacle of the stone land. In contrast, the crawling robot with self-sensing adaptability can accomplish this task easily (Figure 4B and Video S1), and the whole process can be divided into three stages in time sequence. (1) From 0 s to 4 s, the robot passes the rubber land with an actuation frequency of 4 Hz; (2) from 5 s to 11 s, the robot arrives at the intersection of two terrains, and then the robot detects the terrain change and adjusts the actuation frequency from 4 Hz to 1 Hz itself to generate larger output force and stride length according to the result of pre-trained learning model; (3) from 12 s to 22 s, the robot passes the stone land and the PET land smoothly with a self-adjusted frequency of 1 Hz.

The second task contains grass land, plush land, and gravel land in sequential order. As shown in Figure 4C, the unimodal crawling robot takes 28 s to pass all three terrains with a fixed actuation frequency of 1 Hz. In reality, this performance could be a lot worse because we set the frequency at 1 Hz as we already knew it provides the highest efficiency on both grass and gravel lands. We then show that the crawling robot with self-sensing adaptability can finish the same task more efficiently (Figure 4D and Video S2). The whole process can also be divided into three steps in time sequence. (1) From 0 s to 5 s, the robot passes the grass land with an actuation frequency of 1 Hz; (2) from 6 s to 13 s, the robot arrives at the plush land, and then the robot changes its motion mode from 1 Hz to 6 Hz itself for optimal efficiency based on the bionic spine feedback (Figure S12D); (3) from 14 s to 23 s, the robot arrives at the gravel land and again changes its motion mode from 6 Hz back to 1 Hz itself to adapt to the gravel terrain. In this task, the robot equipped with self-adaption crawls over the plush land faster than the unimodal robot by 30%. Thus, the experimental results intuitively show that the proposed self-adaptive crawling robot can actively adapt to the terrain changes and maintain high motion efficiency by autonomous regulation.

Different from other soft robots or living creatures, the self-sensing adaptability of this robot is not dependent on visual feedback, and thus, it has a higher application potential than some living creatures. This spinal sensing strategy makes it possible for the robot to operate all day long in nature regardless of environmental brightness (Figure 4E). To further illustrate this beyond-biological intelligence, we demonstrate the self-adaptive motion of the crawling robot in dazzling lighting conditions and purely dark conditions. Figure 4F shows the crawling robot moving across a multi-terrain pathway under a dazzling lighting condition (Video S3). The monitored spinal feedback signal during this process is shown in Figure S13B, which intuitively reflects the motion states of the crawling robot and the terrain type at different moments. In this experiment, the crawling robot accomplishes self-adaptive motion over different terrains to select an optimal efficiency, with no assistance from vision feedback. Figure 4G also shows that the robot can perform self-adaptive motion for another multi-terrain pathway in purely dark conditions (Figure S13D and Video S4).

An amphibious omnidirectional robot

For broader application scenarios, we construct an amphibious soft robot by combining two spinal actuators as building blocks in parallel like a catamaran (Figure 5A). The two actuators could be driven separately to enable more complex overall motions. Based on a special robotic foot design and a sequential actuation of the left and right legs, the robot can realize nearly omnidirectional motions (more in Text S4). As shown in Figure 5B, when we drive the two robotic legs to move at the same actuation frequency, the motion of the robot is straight movement but a forward direction with a low actuation frequency lower than 8 Hz and a reverse direction with a relatively high actuation frequency higher than 8 Hz. Moreover, the steering of the robot can also be achieved by a different stride length between its two legs. For instance, when the stride length of the right leg is larger than the left one, the robot will turn to the left.

Then, we further demonstrate how this dexterous amphibious robot efficiently accomplishes more complex tasks by self-sensing adaption without any external support. The first scenario is self-adaptive obstacle avoidance. Figure 5C shows the complete avoidance process between the robot and the obstacle (Video S5 and more in Text S5), which can be divided into three stages in time sequence.

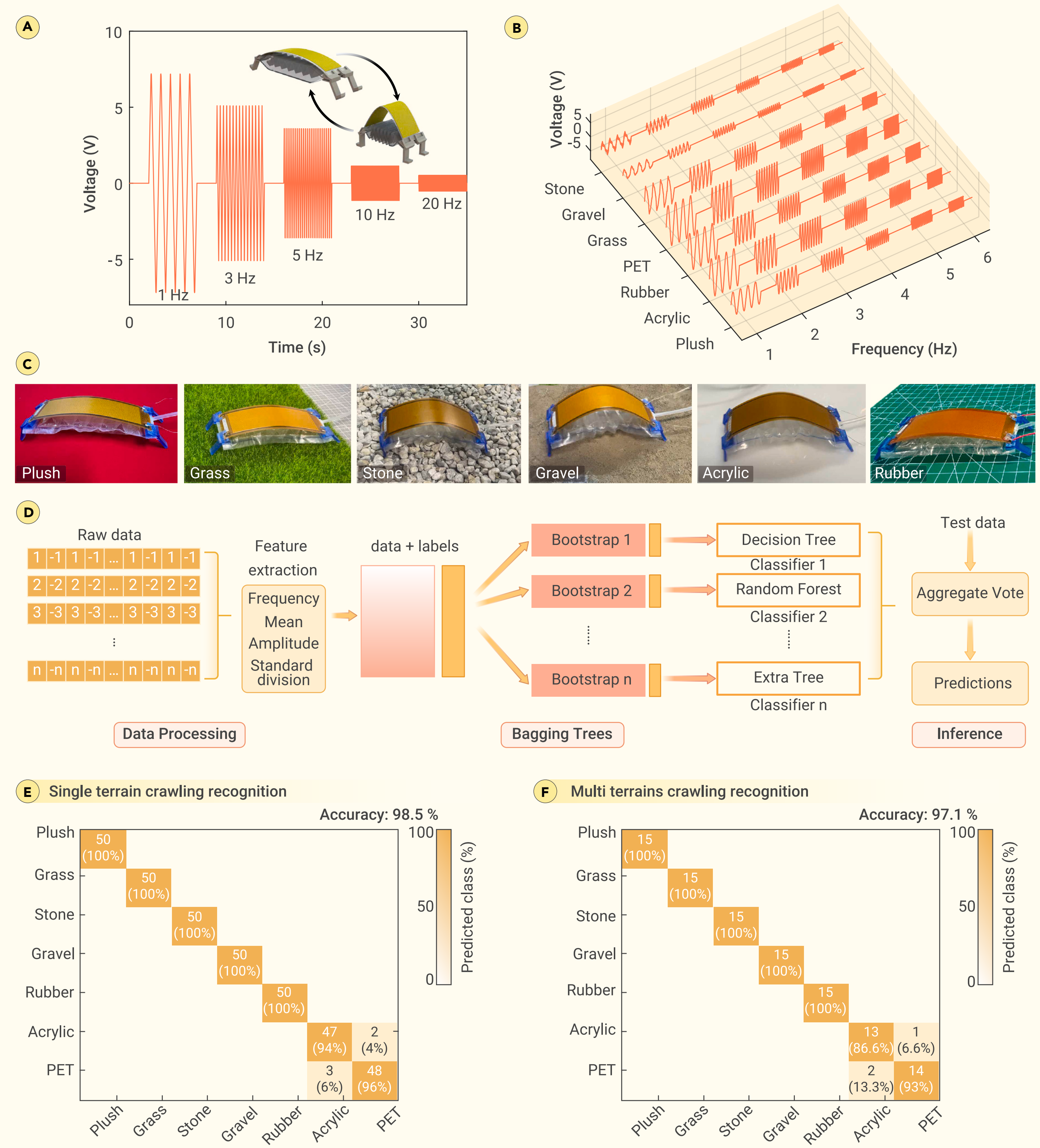


Figure 3. Environmental awareness and recognition by the spinal robot (A) Measured feedback signal by the spine when the robot moves with an increased actuation frequency from 1 Hz to 20 Hz. (B) Measured feedback signal of the spine when the robot crawls on different terrains as a function of actuation frequency. (C) Images of the crawling robot on different terrains, including some relatively smooth terrains, i.e., rubber land, PET land, acrylic land, and plush land, and some uneven terrains, i.e., grass land, gravel land, and stone land. (D) Flow diagram of the machine learning process. (E) Classification confusion matrix of the robotic recognition on single-terrain tasks, with an overall accuracy of 98%. (F) Classification confusion matrix of the robotic recognition on multi-terrain tasks, with an overall accuracy of 97.1%.

(1) The robot moves forward to approach the obstacle and collides with it (from 0 s to 6 s, with an actuation frequency of 1 Hz); (2) the robot detects the obstacle in its path and makes a decision to walk around it, followed by self-adjusted motion state to reverse and move away from the obstacle (from 6 s to 13 s, with an actuation frequency of 8 Hz); and (3) the robot adjusts its pathway and steers to avoid the obstacle (from 13 s to 31 s, with an actuation frequency of 2 Hz). Figure 5C also shows the actuation signals for both of the robotic legs during the whole process. As we mentioned above, this self-adaptive obstacle avoidance is independent from external devices and vision, so it fully demonstrates the active intelligence of the robotic design and is still available in dark environments.

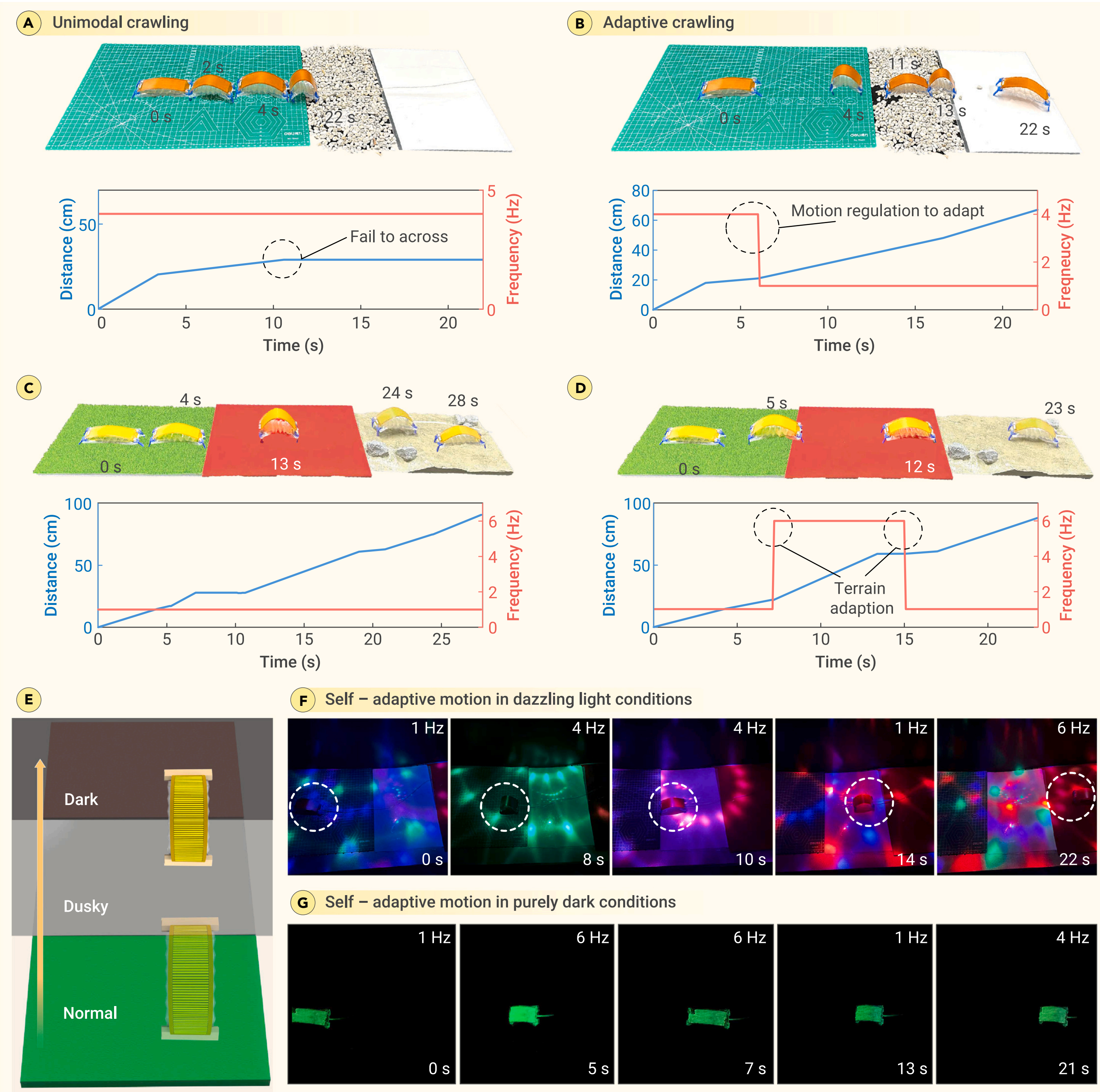


Figure 4. Self-sensing adaption by the spinal robot in multi-terrain tasks (A and C) The unimodal crawling robot moves over two multi-terrains scenarios, as the first scenario (A) contains rubber land, stone land, and PET land, and the second scenario (C) contains grass land, plush land, and gravel land. (B) The spinal crawling robot with self-sensing adaption for the first scenario, overcoming the terrain boundary that the unimodal one failed. (D) The spinal crawling robot for the second scenario, showing it to be 18% faster than the unimodal one. (E) Self-sensing adaption of the spinal robot remains available in dusky and dark environments. (F) Images of the spinal crawling robot performing a self-adaptive multi-terrain-crossing task in dazzling light condition. (G) Images of the spinal crawling robot performing another self-adaptive crossing task in purely dark condition.

The second scenario is self-adaptive amphibious motion, which contains multiple environments at different altitudes, including gravel land, grass land, soil descent, and aquatic area (Video S6). As shown in Figure 5D, we divide the whole self-adaptive motion of the amphibious robot into three stages in time sequences. (1) The robot moves forward and performs self-adaptive multi-terrain transitions between plush land, gravel land, and the grass land (from 0 s to 15 s); (2) the robot undergoes a soil descent and gradually moves closer to the aquatic area (from 16 s to 25 s, with an actuation frequency of 2 Hz); and (3) the robot adapts to the aquatic area to swim efficiently (from 26 s to 33 s, with a self-adjusted frequency of 4 Hz). The self-sensing adaptability has resulted in a 22% improvement in overall time when moving across this land-aquatic path compared to the unimodal robot (Figures S19 and S20).

CONCLUSION

In summary, we have shown that a unibody sensor-actuator-integrated spine provides a potential for high-level intelligence for soft robots. As the core component in our proposal, a piezoelectric macro-fiber composite sheet acts as a bionic spine for both perception and auxiliary actuation, facilitating a synergistic and complementary action between the sensor and actuator. More importantly, we demonstrate that this highly intelligent

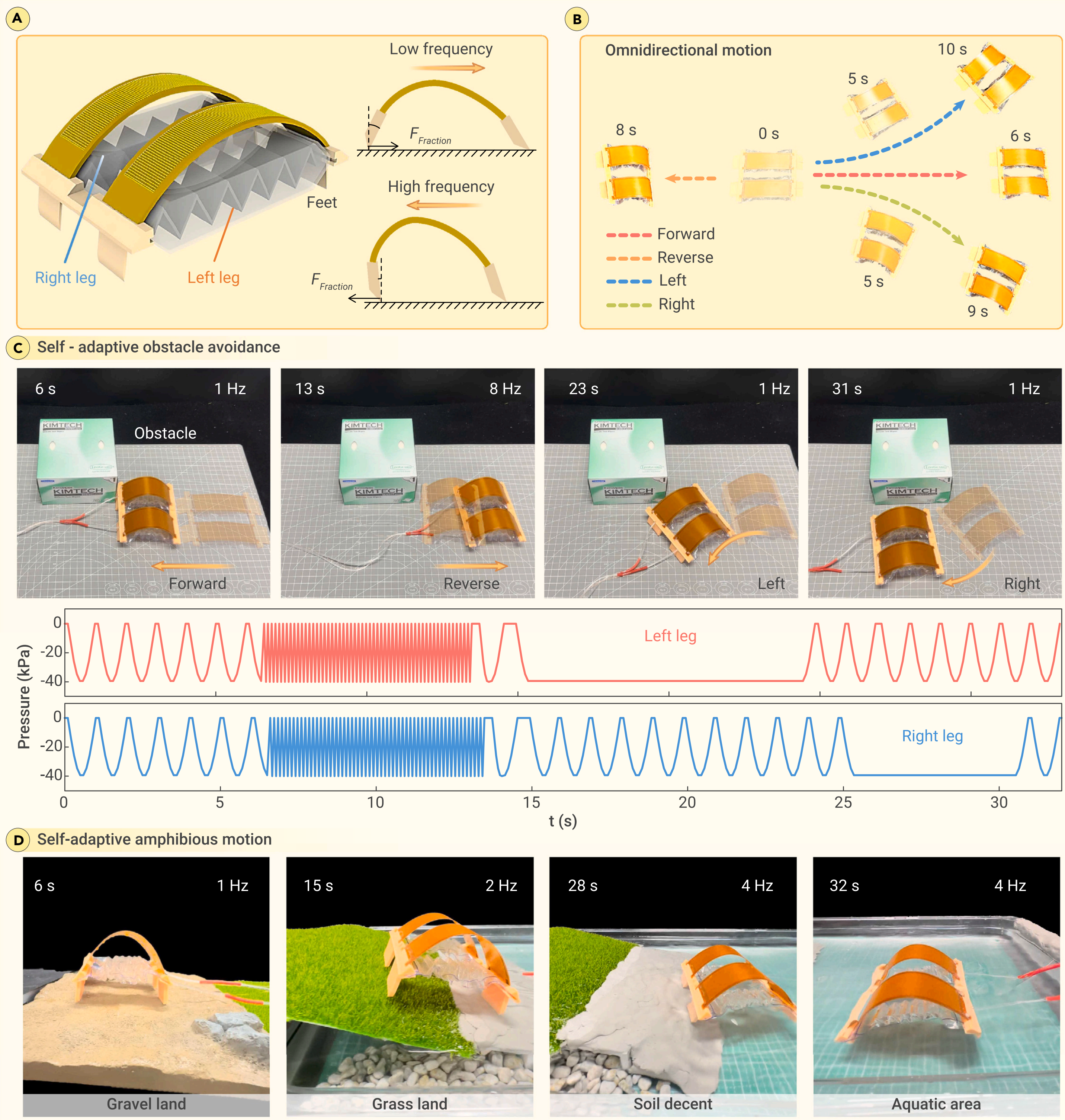


Figure 5. An amphibious and omnidirectional spinal soft robot (A) Design and reversible locomotive mechanism of the robot consisting of twin actuators in parallel. (B) Omnidirectional motions of the robot. (C) Images of the robot performing a self-adaptive obstacle avoidance with self-sensing adaption and dexterous motion. (D) Images of the robot performing multi-terrain crossing and amphibious transition with self-sensing adaption.

soft robot can accomplish beyond-biological applications with vision-free self-sensing adaption, including efficient multi-terrain transition, obstacle avoidance, and amphibious locomotion.

Although there is clearly excellent environmental compatibility and decision-making capacity, the performance of this proposed soft robot still lags behind advanced vertebrate animals, in terms of agility, responsiveness, etc. This is mainly due to the limitations of soft materials and robotic perception systems, causing a worse overall responsive speed compared to vertebrates. Therefore, a dexterous and robust soft actuator with highly sensitive perception is desired to construct a more advanced version based upon this work for broader deployments. Our approach is the first to promote the synergy between sensor and actuator components to implement autonomous adaption, and this could spark future innovations for high-level intelligent soft robots.

MATERIALS AND METHODS

See supplemental information for details.

REFERENCES

1. Rus, D., and Tolley, M.T. (2015). Design, fabrication and control of soft robots. Nature ***521***(7553): 467–475. https://doi.org/10.1038/nature14543.
2. Xiong, J., Chen, J., and Lee, P.S. (2021). Functional Fibers and Fabrics for Soft Robotics, Wearables, and Human-Robot Interface. Adv. Mater. ***33***(19): 2002640. https://doi.org/10.1002/adma.202002640.
3. Han, C., Jeong, Y., Ahn, J., et al. (2023). Recent Advances in Sensor-Actuator Hybrid Soft Systems: Core Advantages, Intelligent Applications, and Future Perspectives. Adv. Sci. ***10***: 2302775. https://doi.org/10.1002/advs.202302775.
4. Li, G., Chen, X., Zhou, F., et al. (2021). Self-powered soft robot in the Mariana Trench. Nature ***591***(7848): 66–71. https://doi.org/10.1038/s41586-020-03153-z.
5. Li, M., Pal, A., Aghakhani, A., et al. (2022). Soft actuators for real-world applications. Nat. Rev. Mater. ***7***(3): 235–249. https://doi.org/10.1038/s41578-021-00389-7.
6. Kim, S., Laschi, C., and Trimmer, B. (2013). Soft robotics: a bioinspired evolution in robotics. Trends Biotechnol. ***31***(5): 287–294. https://doi.org/10.1016/j.tibtech.2013.03.002.
7. Zhang, Y., Zhang, W., Gao, P., et al. (2022). Finger-palm synergistic soft gripper for dynamic capture via energy harvesting and dissipation. Nat. Commun. ***13***(1): 7700. https://doi.org/10.1038/s41467-022-35479-9.
8. Ma, K., Chen, X., Zhang, J., et al. (2023). Inspired by Physical Intelligence of an Elephant Trunk: Biomimetic Soft Robot With Pre-Programmable Localized Stiffness. IEEE Rob. Autom. Lett. ***8***(5): 2898–2905. https://doi.org/10.1109/LRA.2023.3256922.
9. Yao, T., Kos, Ž., Zhang, Q.X., et al. (2022). Nematic Colloidal Micro-Robots as Physically Intelligent Systems. Adv. Funct. Mater. ***32***(44): 2205546. https://doi.org/10.1002/adfm.202205546.
10. Hausladen, M.M., Zhao, B., Kubala, M.S., et al. (2022). Synthetic growth by self-lubricated photopolymerization and extrusion inspired by plants and fungi. Proc. Natl. Acad. Sci. USA ***119***(33): e2201776119. https://doi.org/10.1073/pnas.2201776119.
11. Zhao, Y., Chi, Y., Hong, Y., et al. (2022). Twisting for soft intelligent autonomous robot in unstructured environments. Proc. Natl. Acad. Sci. USA ***119***(22): e2200265119. https://doi.org/10.1073/pnas.2200265119.
12. Sun, J., Lerner, E., Tighe, B., et al. (2023). Embedded shape morphing for morphologically adaptive robots. Nat. Commun. ***14***(1): 6023. https://doi.org/10.1038/s41467-023-41708-6.
13. Baines, R., Patiballa, S.K., Booth, J., et al. (2022). Multi-environment robotic transitions through adaptive morphogenesis. Nature ***610***(7931): 283–289. https://doi.org/10.1038/s41586-022-05188-w.
14. Ilami, M., Bagheri, H., Ahmed, R., et al. (2021). Materials, Actuators, and Sensors for Soft Bioinspired Robots. Adv. Mater. ***33***(19): 2003139. https://doi.org/10.1002/adma.202003139.
15. Gu, G., Zhang, N., Xu, H., et al. (2023). A soft neuroprosthetic hand providing simultaneous myoelectric control and tactile feedback. Nat. Biomed. Eng. ***7***(4): 589–598. https://doi.org/10.1038/s41551-021-00767-0.
16. Dai, J., Li, X.-G., Zhang, T.-Y., et al. (2024). Illuminating a bacterial adaptation mechanism: Infrared-driven cell division in deep-sea hydrothermal vent environments. The Innovation Geoscience ***2***(1): 100050. https://doi.org/10.59717/j.xinn-geo.2024.100050.
17. Tan, D., and Xu, B. (2023). Advanced Interfacial Design for Electronic Skins with Customizable Functionalities and Wearability. Adv. Funct. Mater. ***33***: 2306793. https://doi.org/10.1002/adfm.202306793.
18. Peng, X., Dong, K., Ye, C., et al. (2020). A breathable, biodegradable, antibacterial, and self-powered electronic skin based on all-nanofiber triboelectric nanogenerators. Sci. Adv. ***6***(26): eaba9624. https://doi.org/10.1126/sciadv.aba9624.
19. Zou, Z., Zhu, C., Li, Y., et al. (2018). Rehealable, fully recyclable, and malleable electronic skin enabled by dynamic covalent thermoset nanocomposite. Sci. Adv. ***4***(2): eaaq0508. https://doi.org/10.1126/sciadv.aaq0508.
20. Park, J., Kim, D., and Kim, Y.T. (2021). Soft and transparent triboelectric nanogenerator based E-skin for wearable energy harvesting and pressure sensing. Nanotechnology ***32***(38): 385403. https://doi.org/10.1088/1361-6528/ac0c3f.
21. Guo, N., Han, X., Liu, X., et al. (2023). Autoencoding a Soft Touch to Learn Grasping from On-Land to Underwater. Adv. Intell. Syst. ***6***(3): 2200269. https://doi.org/10.1002/aisy.202300382.
22. Wang, X., Fang, G., Wang, K., et al. (2020). Eye-in-Hand Visual Servoing Enhanced With Sparse Strain Measurement for Soft Continuum Robots. Autom. Lett. ***5***(2): 2161–2168. https://doi.org/10.1109/LRA.2020.2969953.
23. Liu, X., Onal, C.D., and Fu, J. (2023). Reinforcement Learning of CPG-Regulated Locomotion Controller for a Soft Snake Robot. IEEE Trans. Robot. ***39***(5): 3382–3401. https://doi.org/10.1109/TRO.2023.3286046.
24. Hawkes, E.W., Blumenschein, L.H., Greer, J.D., et al. (2017). A soft robot that navigates its environment through growth. Sci. Robot. ***2***(8): eaan3028. https://doi.org/10.1126/scirobotics.aan3028.
25. Huang, C., Xu, T., Liu, J., et al. (2019). Visual Servoing of Miniature Magnetic Film Swimming Robots for 3-D Arbitrary Path Following. IEEE Rob. Autom. Lett. ***4***(4): 4185–4191. https://doi.org/10.1109/LRA.2019.2931234.
26. Xu, F., Wang, H., Liu, Z., et al. (2022). Visual Servoing Pushing Control of the Soft Robot with Active Pushing Force Regulation. Soft Robot. ***9***(4): 690–704. https://doi.org/10.1089/soro.2020.0178.
27. Yang, X., Lan, L., Pan, X., et al. (2023). Bioinspired soft robots based on organic polymer-crystal hybrid materials with response to temperature and humidity. Nat. Commun. ***14***(1): 2287. https://doi.org/10.1038/s41467-023-37964-1.
28. Wang, C., Wu, Y., Dong, X., et al. (2023). In situ sensing physiological properties of biological tissues using wireless miniature soft robots. Sci. Adv. ***9***(23): eadg3988. https://doi.org/10.1126/sciadv.adg3988.
29. Su, Q., Zou, Q., Li, Y., et al. (2021). A stretchable and strain-unperturbed pressure sensor for motion interference–free tactile monitoring on skins. Sci. Adv. ***7***(48): eabi4563. https://doi.org/10.1126/sciadv.abi4563.
30. Lin, W., Zhang, D., Lee, W.W., et al. (2022). Super-resolution wearable electrotactile rendering system. Sci. Adv. ***8***(36): eabp8738. https://doi.org/10.1126/sciadv.abp8738.
31. Li, D., Zhou, J., Yao, K., et al. (2022). Touch IoT enabled by wireless self-sensing and haptic-reproducing electronic skin. Sci. Adv. ***8***(51): eade2450. https://doi.org/10.1126/sciadv.ade2450.
32. Liu, W., Duo, Y., Liu, J., et al. (2022). Touchless interactive teaching of soft robots through flexible bimodal sensory interfaces. Nat. Commun. ***13***(1): 5030. https://doi.org/10.1038/s41467-022-32702-5.
33. Jiang, S., Zhang, T., Zhou, Y., et al. (2023). Wearable ultrasound bioelectronics for healthcare monitoring. Innovation ***4***(4): 100447. https://doi.org/10.1016/j.xinn.2023.100447.
34. Zhang, T., Liu, N., Xu, J., et al. (2023). Flexible electronics for cardiovascular healthcare monitoring. Innovation ***4***(5): 100485. https://doi.org/10.1016/j.xinn.2023.100485.
35. Shahsavan, H., Aghakhani, A., Zeng, H., et al. (2020). Bioinspired underwater locomotion of light-driven liquid crystal gels. Proc. Natl. Acad. Sci. USA ***117***(10): 5125–5133. https://doi.org/10.1073/pnas.1917952117.
36. Truby, R.L., Santina, C.D., and Rus, D. (2020). Distributed Proprioception of 3D Configuration in Soft, Sensorized Robots via Deep Learning. IEEE Rob. Autom. Lett. ***5***(2): 3299–3306. https://doi.org/10.1109/LRA.2020.2976320.
37. Jin, L., Li, Z., Liu, Z., et al. (2022). Flexible unimodal strain sensors for human motion detection and differentiation. npj Flex. Electron. ***6***(1): 74. https://doi.org/10.1038/s41528-022-00205-4.
38. Kim, M., Yu, A., Kim, D., et al. (2023). Multi-Agent Control of Laser-Guided Shape-Memory Alloy Microrobots. Adv. Funct. Mater. ***33***: 2304937. https://doi.org/10.1002/adfm.202304937.
39. Hu, D., Giorgio-Serchi, F., Zhang, S., et al. (2023). Stretchable e-skin and transformer enable high-resolution morphological reconstruction for soft robots. Nat. Mach. Intell. ***5***(3): 261–272. https://doi.org/10.1038/s42256-023-00622-8.
40. Shih, B., Shah, D., Li, J., et al. (2020). Electronic skins and machine learning for intelligent soft robots. Sci. Robot. ***5***(41): eaaz9239. https://doi.org/10.1126/scirobotics.aaz9239.
41. Liu, W., Duo, Y., Chen, X., et al. (2023). An Intelligent Robotic System Capable of Sensing and Describing Objects Based on Bimodal, Self-Powered Flexible Sensors. Adv. Funct. Mater. ***33***(41): 2306368. https://doi.org/10.1002/adfm.202306368.
42. Deng, M., Fan, F., and Wei, X. (2023). Learning-Based Object Recognition via a Eutectogel Electronic Skin Enabled Soft Robotic Gripper. IEEE Rob. Autom. Lett. ***8***(11): 7424–7431. https://doi.org/10.1109/LRA.2023.3316096.
43. Thuruthel, T.G., Shih, B., Laschi, C., et al. (2019). Soft robot perception using embedded soft sensors and recurrent neural networks. Sci. Robot. ***4***(26): eaav1488. https://doi.org/10.1126/scirobotics.aav1488.
44. Jin, T., Sun, Z., Li, L., et al. (2020). Triboelectric nanogenerator sensors for soft robotics aiming at digital twin applications. Nat. Commun. ***11***(1): 5381. https://doi.org/10.1038/s41467-020-19059-3.
45. Xie, M., Zhu, M., Yang, Z., et al. (2021). Flexible self-powered multifunctional sensor for stiffness-tunable soft robotic gripper by multimaterial 3D printing. Nano Energy ***79***: 105438. https://doi.org/10.1016/j.nanoen.2020.105438.
46. Gong, S., Ding, Q., Wu, J., et al. (2022). Bioinspired Multifunctional Mechanoreception of Soft–Rigid Hybrid Actuator Fingers. Adv. Intell. Syst. ***4***(5): 2100242. https://doi.org/10.1002/aisy.202100242.
47. Yan, H., Wang, Y., Shen, W., et al. (2022). Cable-Driven Continuum Robot Perception Using Skin-Like Hydrogel Sensors. Adv. Funct. Mater. ***32***(34): 2203241. https://doi.org/10.1002/adfm.202203241.
48. Jiao, Z., Hu, Z., Shi, Y., et al. (2024). Reprogrammable, intelligent soft origami LEGO coupling actuation, computation, and sensing. Innovation ***5***(1): 100549. https://doi.org/10.1016/j.xinn.2023.100549.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China, China (12102250 and 12032015), the Science and Technology Innovation Action Plan of Shanghai, China (21190760100) the Medical Engineering Interdisciplinary Research Fund for "Star of Jiao Tong University" of Shanghai Jiao Tong University, China, and the Oceanic Interdisciplinary Program of Shanghai Jiao Tong University (SL2023MS006)

AUTHOR CONTRIBUTIONS

S.G. and F.F. initiated the concept with the help of Z.Y., L.S., and W.Z. S.G. and F.F. designed and performed all the experiments under the guidance of Z.Y., L.S., and W.Z., with help from A.L., B.F., and W.L. S.G. wrote the initial manuscript, which was revised by Z.Y. L.S. and W.Z. supervised the project and provided funding. All authors contributed to the discussion and writing of this manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

It can be found online at https://doi.org/10.1016/j.xinn.2024.100640.

LEAD CONTACT WEBSITE

https://me.sjtu.edu.cn/teacher_directory1/zhangwenming.html.

The Innovation, Volume *5*

Supplemental Information

An intelligent spinal soft robot with self-sensing adaptability

Shoulu Gong, Fuyi Fang, Zhiran Yi, Bohan Feng, Anyu Li, Wenbo Li, Lei Shao, and Wenming Zhang

Supplemental Information

An intelligent spinal soft robot with self-sensing adaptability

DOI:

Shoulu Gong, Fuyi Fang, Zhiran Yi, Bohan Feng, Anyu Li, Wenbo Li, Lei Shao, and Wenming Zhang

Table of Contents

Materials and methods

Fabrication of the origami pneumatic actuator

The robot body is an origami pneumatic actuator, which contains a PET sheet, two TPU membranes and a plastic tube. The whole fabrication process of the origami actuator is hand made within 10 minutes, and can be separated into 6 steps (Fig. S1). i) Cut a piece of PET sheets into a size of 20×130 mm. ii) Fold the PET sheet into polygonal patterns as the skeleton of the origami pneumatic actuator, while each part of the polygonal patterns is 20×10 mm in size. iii) Cut two pieces of TPU membranes into size of 40×150 mm. iv) Seal three edges of the TPU membrane like a pocket (Deli Group 2133, sealing machine). v) Place the prepared PET sheet into the TPU pocket. vi) Insert a plastic tube into the TPU pocket and seal the last edge, and then the origami pneumatic actuator is finished. As for the robotic feet with mounting slots, they were first printed with a resin 3D printer (Formlabs, Form 3+) and then glued to the origami pneumatic actuator. Afterwards, the piezoelectric macro-fiber composite sheet was assembled on the origami actuator via the mounting slots on the robotic feet to form the complete structure of the robot.

Characterization and actuation of the pneumatic origami actuator

The output force of the origami pneumatic actuator under different actuation modes is monitored by a force meter (HP-2, Wenzhou Haibao Instrument Co., Ltd.). The stride length is recorded by a high-speed camera (Canon, EOS R3). The variable frequency in the actuation signal is generated by a miniature air pump in combination with a relay control circuit.

Actuation and characterization of the piezoelectric macro-fiber composite sheet

The actuation signal of the robot spine is generated by a digital signal generator (UNI-T, UTG932) and amplified by an amplifier circuit (HVA1500/50, COREMORROW) to actuate the spine for auxiliary actuation. During the sensing performance analysis stage, the feedback signal is monitored by an oscilloscope. During the active environment adaption process, the feedback piezoelectric signal is captured by the IO port on an Arduino Mega board through a precise charge amplifier (Vk10X, for high precision and ultra-low noise voltage-to-charge conversion) and was subsequently analyzed and processed in MATLAB.

Machine learning modeling process

To decouple the motion states and the environmental interaction information, machine learning techniques are utilized to process the feedback signal for further robot self-learning and adaptation. The entire modeling and prediction process is shown in Fig. 3d. Initial raw data was processed to extract key statistical features such as frequency, mean, amplitude, and standard deviation. These statistical features were then input into our decision tree classifiers to form a predictive model. Recognizing patterns in the data, the classifiers converted the statistics into executable insights. In addition, we enhanced this approach by bagging trees. This involves creating multiple decision tree models, with each trained on a random subset of the data. Their predictions were aggregated, and the majority vote determined the final prediction, thereby reducing the risk of overfitting and boosting classification accuracy.

Text 1. Modeling and characterization of the spinal robot

The pneumatic origami actuator is ultra-light and of high robustness, with common materials and fast fabrication. As is shown in Fig. S1, the origami pneumatic actuator is manufactured with PET sheet, TPU membranes and plastic tube. The whole fabrication process can be operated by hand within 10 minutes, which can be separated into 6 steps: i) cut a piece of PET sheet with size of 20×130 mm; ii) fold the PET sheet as the skeleton of the origami pneumatic actuator, each part is 20×10 mm in size; iii) cut two piece of TPU membrane with size of 40×150 mm; iv) seal three side of TPU membrane and left one of the short sides like a pocket (the sealing machine is "plastic sealer SF-400"); v) put the prepared PET sheet into the TPU pocket; vi) insert a plastic tube into the TPU pocket and seal the last side, the origami pneumatic actuator is finished.

To improve the efficiency of manufacturing, the PET sheets and TPU membranes can all be fabricated by laser cutting instead of handmaking. The sealing process can also be replaced by a special sealing mold, which is more suitable for mass manufacturing.

As for the Soft Intelligent Crawling Robot, it is composed of an origami pneumatic actuator, a MFC piezoelectric macro-fiber composite sheet and four 3D printed feet (*Fig 1b*). The feet are fabricated by a resin 3D printer (Formlabs, Form 3+), with mounting slots on them. The lower mounting slot is fixed on the corners of the prepared origami pneumatic actuator with glue (Loctite 416), then the piezoelectric macro-fiber composite sheet is pre-bent and inserted into the upper mounting slots in the same way (fixed with Loctite 416 glue). The gas tube of the origami pneumatic actuator and wires of the piezoelectric macro-fiber composite sheet are respectively connected to vacuum pump and H-bridge circuit.

(1) Modeling process of the pneumatic origami actuator

The force of artificial muscles is provided by the difference of internal and external pressure, and the internal skeleton not only supports the shape of the artificial muscles, but also provides elasticity[48]. Therefore, the elasticity of the skeleton should be subtracted when calculating the total contraction force. The parameters are shown in Fig. S2, and the derivation process is as follows:

Calculate the total contraction force caused by pressure difference using the principle of virtual work:

$$F(\theta) \times 2\delta L(\theta) = -\Delta p \times \delta V(\theta) \tag{S1}$$

$$F(\theta) = -\frac{\Delta p}{2}\left[\frac{\delta V(\theta)/\delta\theta}{\delta L(\theta)/\delta\theta}\right] \tag{S2}$$

In which $L(\theta) = Dsin\theta$, $\frac{\delta L(\theta)}{\delta\theta} = Dcos\theta$

Simplify the model of artificial muscles, with parameters shown in *Fig S2c*

$$V(\theta) = V_{void} - V_{triangle} \tag{S3}$$

$$V_{void} = W \times A_{void} = WLH = WD^2 sin\theta cos\theta \tag{S4}$$

$$V_{triangle} = W \times A_{triangle} = WLh = WDsin\theta[s_0^2 - (Dsin\theta)^2]^{\frac{1}{2}} \tag{S5}$$

$$s_0 = \frac{1}{2}\left[(L_0^2 + 4h_0^2)^{\frac{1}{2}} + \frac{L_0^2}{2h_0} sinh^{-1}\left(\frac{2h_0}{L_0}\right)\right] \tag{S6}$$

Let $\mu = s_0^2 - D^2 sin^2\theta$, Eq (S2) can be expressed as

$$\frac{\delta V(\theta)}{\delta\theta} = WD\left[D(cos^2\theta - sin^2\theta) - \mu^{\frac{1}{2}}cos\theta + \mu^{-\frac{1}{2}}D^2 sin^2\theta cos\theta\right] \tag{S7}$$

$$F(\theta) = -\frac{\Delta pW}{2}\left[\frac{D(cos^2\theta - sin^2\theta) - \mu^{\frac{1}{2}}cos\theta + \mu^{-\frac{1}{2}}D^2 sin^2\theta cos\theta}{cos\theta}\right] \tag{S8}$$

The skeleton of the pneumatic origami actuator can be simplified as a spring, and its elasticity is expressed as

$$F_{elastic}(\theta) = k_1\Delta L = k_1(L_0 - Dsin\theta) \tag{S9}$$

The elastic coefficient in which is $k_1 = \frac{EWt^3}{4D^3}$

Thus, the total output force and total contraction are

$$F_{output}(\theta) = F(\theta) - F_{elastic}(\theta) \tag{S10}$$

$$C(\theta) = N \times 2[L_0 - L(\theta)] = 2ND(sin\theta_0 - sin\theta) \tag{S11}$$

(2) The relationship between the vacuum pressure and contraction rate of the pneumatic actuator with different load and skeleton thickness

To obtain the properties of the origami pneumatic actuator, we tested how vacuum pressure, thickness of the skeleton and load influence the contraction rate.

We fixed one end of origami pneumatic actuator (the end close to the gas tube), and applied vacuum pressure to the actuator. The length of the actuator changed along with the vacuum pressure, and different thickness of the skeleton also influence the contraction rate. As is shown in Fig. S3b, the contraction rate is up to 50% under vacuum pressure of 80 kPa (skeleton of 0.1 mm PET and skin of 0.01 mm TPU). As the thickness of the skeleton increased, the maximum contraction rate decreased significantly from 50% to 32% with thickness from 0.1 mm to 0.3 mm under -80 kPa (further experiments indicated that the contraction rate is not sensitive with the plastic type of the skeleton).

To test the influence of the load, we suspended the actuator (0.1 mm PET skeleton and 0.01 mm Bopp skin) vertically and hung weights under it with magnets (the weight of the magnets is 15 grams totally and has been contained in), as shown in Fig. S3a and Fig. S3c. Under Vacuum pressure of 80 kPa, the contraction rate did not reduce apparently (from 57% to 55%) as the load increased to 215 grams. However, under vacuum pressure of 20 kPa, the contraction rate reduced rapidly (from 48% to 21%) as the load increased.

Considering the performance of the actuator and efficiency of the vacuum pump, we finally chose -50 kPa as the actuating pressure.

(3) Releasing force of the bent piezoelectric macro-fiber composite sheet

The piezoelectric macro-fiber composite sheet ($length = 2l, width = b, thickness = 2h$) acts as a spine when the robot is crawling. As the robot "tissue" contracts, the sheet turns into a bending beam. Because the bending beam is symmetrical, we choose half of the beam to analysis (Fig. S4).

The bending beam curvature formula is:

$$\frac{1}{\rho} = \frac{M}{E_s I} \tag{S12}$$

Where $M = F_{release} y$

With the definition of curvature $\frac{1}{\rho} = \frac{d\theta}{ds}$, Eq. (12) can be expressed as

$$\frac{d\theta}{ds} = \frac{F_{release} y}{E_s I} \tag{S13}$$

Derive both sides of Eq. (S13) with respect to s, and consider $\frac{dy}{ds} = sin\theta$, a differential equation can be acquired

$$\frac{d^2\theta}{ds^2} = \frac{F_{release}}{E_s I} sin\theta \tag{S14}$$

Consider the boundary conditions that

The free-end: $\theta|_{s=0} = \partial$, $\frac{d\theta}{ds}|_{s=0} = 0$

The middle point: $\theta|_{s=l} = 0$

To solve the Eq. (S14), the relationship between F and ∂ is emerging

$$F_{release} = \frac{E_s I}{4l^2} \left[\int_0^{\partial} \frac{d\theta}{\sqrt{sin^2 \frac{\partial}{2} - sin^2 \frac{\theta}{2}}} \right]^2 \tag{S15}$$

I is the moment of inertia, that is $I = \frac{1}{12} bh^3$, so Eq. (S15) can be further written as

$$F_{release} = \frac{E_s bh^3}{48l^2} \left[\int_0^{\partial} \frac{d\theta}{\sqrt{sin^2 \frac{\partial}{2} - sin^2 \frac{\theta}{2}}} \right]^2 \tag{S16}$$

Transform Eq. (S16) to the complete elliptic integral

$$F_{release} = \frac{E_s bh^3}{12l^2} \left[\int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1 - sin^2 \frac{\partial}{2} sin^2 \varphi}} \right]^2 \tag{S17}$$

The Young's modulus E_s, width b, half thickness h and half length a are all constants, so we use K_1 to substitute them

$$F_{release} = K_1 \left[\int_0^{\frac{\pi}{2}} \frac{d\varphi}{\sqrt{1 - sin^2 \frac{\partial}{2} sin^2 \varphi}} \right]^2 \tag{S18}$$

The length of x_a and can y_a can also be expressed as

$$x_a = \frac{2}{\sqrt{F_{release}/EI}} \int_0^{\frac{\pi}{2}} \sqrt{1 - sin^2\frac{\partial}{2}sin^2\varphi d\varphi} - l \tag{S19}$$

$$y_a = \frac{sin\frac{\partial}{2}}{\sqrt{F_{release}/EI}} \tag{S20}$$

Thus, the elastic force of the pneumatic actuator skeleton Eq. (S9) can be further expressed with x_a

$$F_{elastic} = k_1\Delta L = k_1\left(L_0 - \frac{2x_a}{n}\right) = C_1 - C_2x_a \tag{S21}$$

In which, n is the buckling number of the pneumatic actuator skeleton.

(4) Auxiliary force of the piezoelectric macro-fiber composite sheet

The piezoelectric macro-fiber composite sheet we use is P1 type, whose electrodes are distributed along the length of the fiber, which means that the P1 type is polarized along the length of the fiber. The strain of the fiber can be expressed as[49]

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \end{bmatrix} = \begin{bmatrix} s_{11}^E & s_{13}^E \\ s_{13}^E & s_{33}^E \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \end{bmatrix} + \begin{bmatrix} d_{31} \\ d_{33} \end{bmatrix} E_z \tag{S22}$$

s^E is the flexibility matrix of the piezoelectric macro-fiber composite sheet, d is the piezoelectric strain constant, and E_z is the electric intensity. Express the stress by the strain

$$\begin{bmatrix} \sigma_x \\ \sigma_y \end{bmatrix} = \begin{bmatrix} Q_{11}^E & Q_{13}^E \\ Q_{13}^E & Q_{33}^E \end{bmatrix} \begin{bmatrix} \varepsilon_x - d_{31}E_z \\ \varepsilon_y - d_{33}E_z \end{bmatrix} \tag{S23}$$

The Q^E is the transposition of s^E, in which

$$Q_{11}^E = \frac{s_{33}^E}{s_{11}^E s_{33}^E - (s_{13}^E)^2},\ Q_{13}^E = \frac{-s_{13}^E}{s_{11}^E s_{33}^E - (s_{13}^E)^2},\ Q_{33}^E = \frac{s_{11}^E}{s_{11}^E s_{33}^E - (s_{13}^E)^2}$$

The strain can also be defined as

$$\begin{cases} \varepsilon_x = \dfrac{z}{\rho_x} = \dfrac{z}{E_sI_x/M_x} = \dfrac{zM_x}{E_sI_x} = \dfrac{zM_x}{E_s\dfrac{1}{12}b(2h)^3} = \dfrac{3zM_x}{2E_sbh^3} \\ \varepsilon_y = \dfrac{z}{\rho_y} = \dfrac{z}{E_sI_y/M_y} = \dfrac{zM_y}{E_sI_y} = \dfrac{zM_y}{E_s\dfrac{1}{12}2l(2h)^3} = \dfrac{3zM_y}{4E_slh^3} \end{cases} \tag{S24}$$

The driving bending moment of the piezoelectric macro-fiber composite sheet is

$$\begin{cases} M_x = 2l\int_0^{2h}(-\sigma_x)zdz = 2l\int_0^{2h}\left[Q_{11}^E(d_{11}E_z-\varepsilon_x)+Q_{13}^E\left(d_{33}E_z-\varepsilon_y\right)\right]zdz \\ M_y = b\int_0^{2h}\left(-\sigma_y\right)zdz = b\int_0^{2h}\left[Q_{13}^E(d_{31}E_z-\varepsilon_x)+Q_{33}^E\left(d_{33}E_z-\varepsilon_y\right)\right]zdz \end{cases} \tag{S25}$$

Consider Eq. (S25) with Eq. (S23) and Eq. (S24), and integrate with respect of z

$$\begin{cases} \left(\dfrac{8lQ_{11}^E}{E_s b}+1\right)M_x+\dfrac{4Q_{13}^E}{E_s}M_y = 4lE_z h^2(Q_{11}^E d_{31}+Q_{13}^E d_{33}) \\ \dfrac{4Q_{13}^E}{E_s}M_x+\left(\dfrac{2bQ_{33}^E}{E_s l}+1\right)M_y = 2bE_z h^2(Q_{13}^E d_{31}+Q_{33}^E d_{33}) \end{cases} \tag{S26}$$

Solve Eq. (S26) to get M_y

$$M_y = \frac{(8lQ_{11}^E+E_s b)(bQ_{13}^E d_{31}+bQ_{33}^E d_{33})-8bQ_{13}^E(lQ_{11}^E d_{31}+lQ_{13}^E d_{33})}{(8lQ_{11}^E+2lE_s)(4bQ_{33}^E+bE_s)-32lb{Q_{13}^E}^2}\cdot 4lh^2E_sE_z \tag{S27}$$

The moment M_y is generated by $F_{auxiliary}$, thus:

$$y_a F_{auxiliary} + x_a\frac{F_{auxiliary}}{tan\partial} = M_y$$

$$F_{auxiliary} = \frac{M_y}{y_a+\dfrac{x_a}{tan\partial}} \tag{S28}$$

The size, Young's modulus, flexibility and piezoelectric strain constant of the MFC are all constants, and the electric intensity $E_z \propto V$, so Eq. (S25) can be expressed as

$$F_{auxiliary} = K_2\frac{V}{y_a+\dfrac{x_a}{tan\partial}} \tag{S29}$$

(5) Demonstration of the movement process of the crawling robot

As shown in Figure S6, the soft crawling robot is consisting of a pneumatic origami actuator, a bionic spine and two printed feet of resin material, and its maximum ability to cross obstacles is about 7 *mm*. During crawling process, the robot's motion can be categorized into two states. i) Contraction state: when a negative pressure is applied, the artificial muscle will contract and the deflection of the bionic spine increases. Due to the anisotropic friction between the foot and the contact surface, i.e., the friction on the foot sliding forward is less than the friction on the foot sliding backward, the crawling robot will move forward. ii) Release state: when the negative pressure is withdrawn, and an auxiliary voltage excitation is applied to the bionic spine to act as the auxiliary actuation, the crawling robot will revert to release state. Currently, the overall position of the crawling robot moves forward, and the distance moved is called the stride length.

Text 2. Perception of the spinal robot

The perception capability of the crawling robot relies on converse piezoelectricity of the bionic spine, and output voltage is directly related to the spinal deformation. Regarding to the proprioception capability of the crawling robot, its motion frequency is visualized in the frequency of feedback signals. More features are valuable for further analysis, for example, larger deformation will cause larger voltage output, which is the vital basis for crawling robot shape reconstruction. As shown in Fig. S8a (i), when the crawling robot is at the initial state, we define the robot length as L_1, the length of the bionic spine as $2l$, the bending angle of the bionic spine as ∂_1, and the equivalent radius of the bionic spine as R_1. The relationship between these parameters can be expressed as:

$$R_1 = \frac{2l}{\partial_1} \tag{S30}$$

$$L_1 = 2R_1 sin\left(\frac{\partial_1}{2}\right) \tag{S31}$$

When the crawling robot moves from the initial state to the contraction state shown in Fig S8a (ii), the novel geometrical relationship between these parameters can be expressed as:

$$R_2 = \frac{2l}{\partial_2} \tag{S32}$$

$$L_2 = 2R_2 sin\left(\frac{\partial_2}{2}\right) \tag{S33}$$

And the contraction rate of the crawling robot can be calculated as:

$$Rate_{contraction} = \frac{L_1 - L_2}{L_1} \tag{S34}$$

Hence, based on the bionic spine feedback signal, we obtain all geometrical parameters to reconstruct the shape of the crawling robot, including the spine deflection, the robot length and so on. And the curvature (bending angle) of the bionic spine directly corresponds to the amplitude of the output voltage. Fig. S8b shows the measured amplitude of output voltage as a function of robot length L. The experimental results show a high agreement with the theoretical analysis. On the one hand, with the increase of the driving frequency, the stride length of the robot decreases (the contraction rate of the robot decreases); On the other hand, the decreased contraction rates lead to a decrease of the bionic spine deformation, causing a decrease of the amplitude of the output voltage (Figure 4a).

Regrading to the exteroception capability of the crawling robot to environmental interactions. When the crawling robot moves on different terrains, in addition to the actuation frequency, the stride length is also directly related to the surface roughness and morphological characteristics of the contact terrains. These differences lead to changes in the features of bionic spine feedback signals, including the amplitude, waveform, etc. For example, the larger surface roughness will cause more resistance to the motion of robots, which leads to changes in the robot stride length and shows up in the amplitude of bionic spine feedback signals. Fig. S9 shows the monitored output voltage amplitude when the crawling robot is moving on different terrains with different actuation frequencies, showing that the crawling robot (at same actuation frequency) usually show smaller stride length when moving on terrains with large roughness (e.g., plush plane and gravel plane, etc.) compared to smooth planes (e.g., PET plane and acrylic plane, etc.).

The waveform of the feedback signals also reflects the topographic information of the contact terrains. For instance, Fig. S10a show the monitored feedback signal of the crawling robot moving on Rubber land, Acrylic land, PET land and Plush land with an actuation frequency of 1 Hz. All four terrains are flat, so the feedback signal is relatively smooth. However, when the crawling robot is moving on uneven ground, the waveform of the e-skin feedback signal will change. Fig. S10b shows the monitored feedback signal of the crawling robot moving on Grass land with an actuation frequency of 1 Hz and the signal is with a slight sawtooth, mainly due to obstacles and surface bumps in the contact terrain, etc. And as the contact terrains be more rugged and complex (e.g., Gravel land and Stone land), this variation in the waveform of the e-skin feedback signal will be more pronounced (shown in Fig. S10c and S10d).

Text 3. Isolation of bionic spine sensing and actuation signals

In this research, we utilized a bridge circuit[49]. This classic circuit is suitable for full piezoelectric set of equations, i.e., the piezoelectric sheet is subjected to both electric field and strain. This circuit has been widely used in piezoelectric valve[50], piezoelectric clamp[51] and piezoelectric pump[52].

In this spinal crawling robot, the bionic spine acts as both sensing and actuation elements. As shown in Figure 1b, V_c is the actuation voltage of the bionic spine and V_{sen} is the feedback voltage of the bionic spine. To eliminate the crosstalk between these two voltage signals and achieve the sensor-actuator integration. A bridge circuit is utilized to construct the self-sensing actuator. C_R is the equivalent binding capacitance of the MFC piezoelectric sheet, the MFC piezoelectric sheet acts as a bridge arm, C_P is the reference capacitance, Z_1 and Z_2 are series impedances. If the bridge circuit is balanced, the feedback sensing voltage V_{sen} is not related to the actuation voltage V_c.

When the series impedances Z_1 and Z_2 are capacitance - capacitance, when $C_R\ C_1 = C_p\ C_2$, the bridge circuit is balanced, and the output U_s can be expressed as:

$$U_{sen} = \frac{C_R}{C_2 + C_R} U_c = \frac{Q}{C_2 + C_R} \tag{S35}$$

And when the series impedances Z_1 and Z_2 are resistance - resistance, when $C_R\ R_2 = C_p\ R_1$, the bridge circuit is balanced, and the output U_s can be expressed as:

$$U_{sen} \approx -I_{sens} R_2 \tag{S36}$$

For this crawling robot, the equivalent capacitance of the MFC sheet is 7 nF, and we choose two capacitances as series impedances Z_1 and Z_2. The corresponding capacitances are C_1 = 330 nF, C_p = 10 nF, C_2 = 231 nF.

<u>Text 4. Omnidirectional motion of the spinal amphibious robot</u>

As shown in Figure 5a, we design another spinal amphibious soft robot, which contains two actuators and designed robot feet, with capabilities of omnidirectional motion.

As shown in Figure 5b. The turning motion are achieved by actuating one of the pneumatic origami actuators. Because of the deformation asymmetry, the robot bends and turns to the direction in which the actuator contracts more. For example, when driving the right one and keeping the left one under vacuum pressure, the robot bends to the left and moves leftward; on the contrary, when driving the left one and keeping the right one under vacuum pressure, the robot moves rightward.

The forward motion and reverse motion of the crawling robot are achieved by the designed feet and specific actuating frequency. Fig. S14 shows the diagram of the feet and three different gaits of the crawling robot under different actuating frequency intervals. Both front feet and rear feet are composed of a pointed tip and a flat side, and the pointed tip sticks to the front at the initial state. It should be noticed that the lifting angle of the rear feet is always larger than the front feet, as the gas tube and signal wires limit the front feet to lift. When applying a low actuating frequency (1~2 Hz) to the actuators, the front feet firstly stick to the terrain with the pointed tip, and as the actuators contract, the rear feet are dragged forward and lifted. During the restoring period, the rear feet of the robot keep anteversion, i.e., they act as an anchor to push the robot forward. And because the robot has enough time to deform and restore under low actuating frequency, the stride length of the crawling robot is relatively large. When applying a middle actuating frequency (2~8 Hz), the rear foot of the robot still keeps anteversion and push the crawling robot forward, but as the actuating period becomes shorter, the robot cannot restore adequately, the stride length becomes smaller. When the actuating frequency is larger than 8 Hz, the actuators keep a much higher contraction rate, making the rear feet to turn tilt-back, and act as an anchor during the contracting period, at the same time, the front feet are dragged back because the flat side allows smaller friction. During the restoring period, the front feet anchor to the terrain instead, and the rear feet move backward, the robot is pulled backward. With these motion modes, the intelligent crawling robot is able to avoid obstacles on the path. Figure S15 shows the measured crawling and rotation speed of the omnidirectional robot as a function of actuation frequencies on the rubber land.

Fig. S16 shows the active obstacle avoiding process of the spinal robot. The robot moves forward on the cutting board with 1 Hz firstly, as is blocked, the robot identifies the obstacle with feedback signal, and adjusts the actuating frequency to 8 Hz correspondingly, driving it backward. This capability promises the crawling robot to achieve more complex tasks and adapt to unstructured environment.

Text 5. Demonstration of the self-adaptive obstacle avoidance

As shown in Figure S17, the omnidirectional crawling robot can realize self-adaptive obstacle avoidance. Similarly, the obstacle avoidance process of the robot also can be divided into three stages. During the sensing stage: the motion states of the robot and its interaction with the environment (both contacted terrains and obstacles). At the avoidance stage: features of the monitored feedback signals will be utilized for terrains recognition based on a custom machine learning model. Collisions between robots and obstacles often cause large fluctuations in the feedback signal of the bionic spine, which can be utilized as a basis for judgment. In actuation stage, the motion direction of the robot will be self-regulated to avoid the obstacle.

Taking the self-adaptive obstacle avoidance task shown in Figure 5c, Figure S18 shows the monitored feedback signal from e-skin as a function of time during the whole motion process. The robot's right foot touched the obstacle at 6s, and it can be observed that the feedback signals of both the left and right bionic spine are mutated, and the mutation is more pronounced on the right side (the blue line). By capturing the change of features in the feedback signal, the robot can achieve obstacle detection on any terrain. After that, the motion direction of the robot will change from forward to backward to move away from the obstacle. Then, the robot will move in the opposite direction of the obstacle.

Supplementary Table 1

Motion efficiency improvement of the self-adaption robot on different terrains

	Grass	**Plush**	**Rubber**	**Gravel**	**Stone**	**PET**	**Acrylic**
Max speed (mm/s)	48	96	60	52.5	25	54	48
Min speed (mm/s)	28.33	55	43	22	0	30	27
Enhancement (%)	69.4	74.5	39.5	238	∞	80	77.8

Supplementary Table2

Comparison of soft sensors

	Sensitivity (Output / Input)	**Signal-noise ratio**	**Robustness**	**Stability**
Piezoelectric[53,54]	5	70 dB	High	High
Hydrogel[55-57]	0.95	59.4 dB	Fragile	Low
Liquid metal[58]	0.2	37.4 dB	Fragile	High
Magnetic[59-61]	5×10^{-5}	45 dB	High	High

Supplementary Table 3

Comparison of soft artificial muscles

	Fabrication		**Actuating require-ment**	**Maximum deformation ratio**	**Maximum load-weight ratio**	**Maximum actuating frequency**
	Materials	**Time cost**				
Origami artificial muscle[62]	PET sheet; TPU membrane; TPU tube.	<10 min	-80 kPa	90%	1200	20 Hz
Networks[63]	Silicone; Soft tube.	4-6 h	80 kPa	100%	13	5 Hz
McKibben muscles[64,65]	Inner chamber; Mesh sleeve; Ties.	10-30 min	500 kPa	30%	5000	10 Hz
SMA[66]	Alloys.	>1 h	50-300 ℃	10%	20 N/mm^2	Typical low frequency.
HASEL[67-69]	PET membrane; Electrodes; Oil.	10-30 min	8 kV	10%	200	50 Hz
DEA[70]	Dielectric elastomer; Electrodes.	10-30 min	1 kV	10%	200	50 Hz

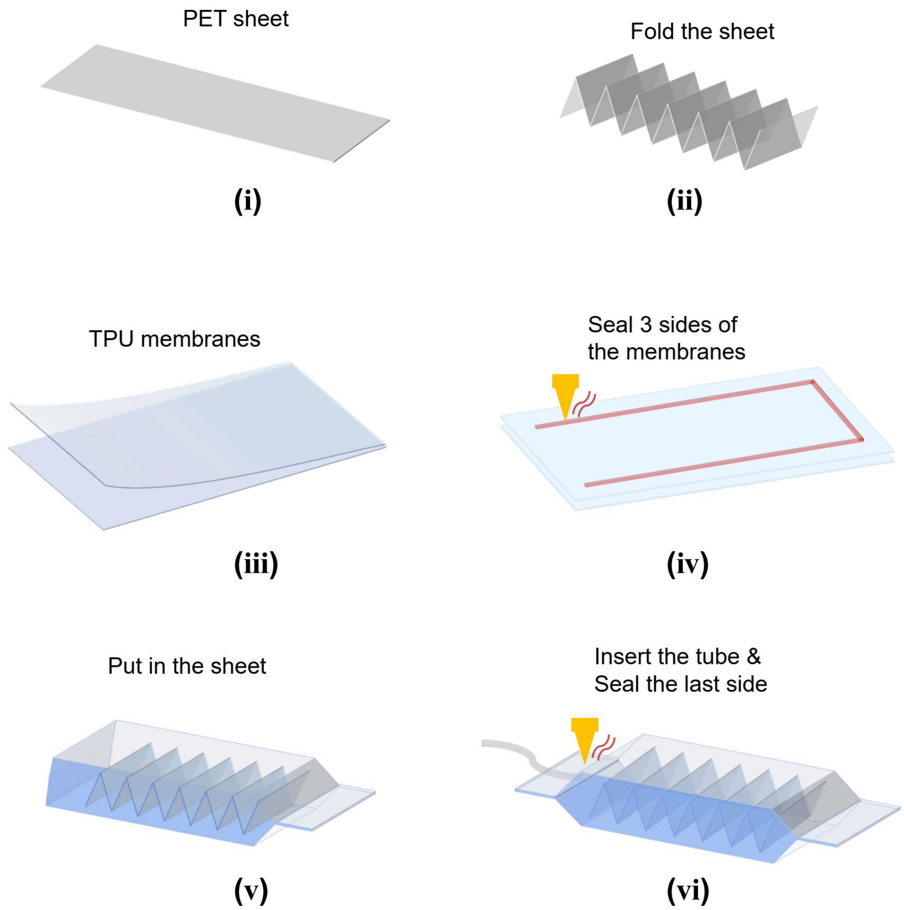


Figure S1. Fabrication detail of the pneumatic origami actuator. i) Cut a piece of PET sheet with size of 20×130 mm. ii) Fold the PET sheet as the skeleton of the origami pneumatic actuator, each part is 20×10 mm in size. iii) Cut two pieces of TPU membrane with size of 40×150 mm. iv) Seal three sides of TPU membrane and left one of the short sides like a pocket. v) Put the prepared PET skeleton into the TPU pocket. vi) Insert a plastic tube into the TPU pocket and seal the last side.

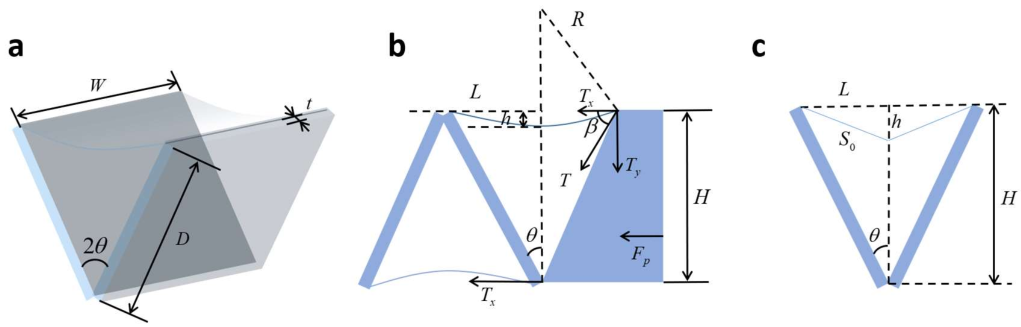


Figure S2. Modeling and characterization of the pneumatic origami actuator. (a) Characteristics of a unit in the pneumatic origami actuator, including the width of the skeleton W, the length of a skeleton unit D, the thickness of the skeleton t and the included angle of a skeleton unit 2θ. (b) Characteristics of the pneumatic origami actuator's cross section, including the height of the actuator H, the depressed distance of the membrane h, the half span of the unit L, the tractive force of the membrane T and the pressure by the environment F_p. (c) Simplified model of a unit in the pneumatic origami actuator, S_0 is the half length of the membrane between two pieces of skeleton.

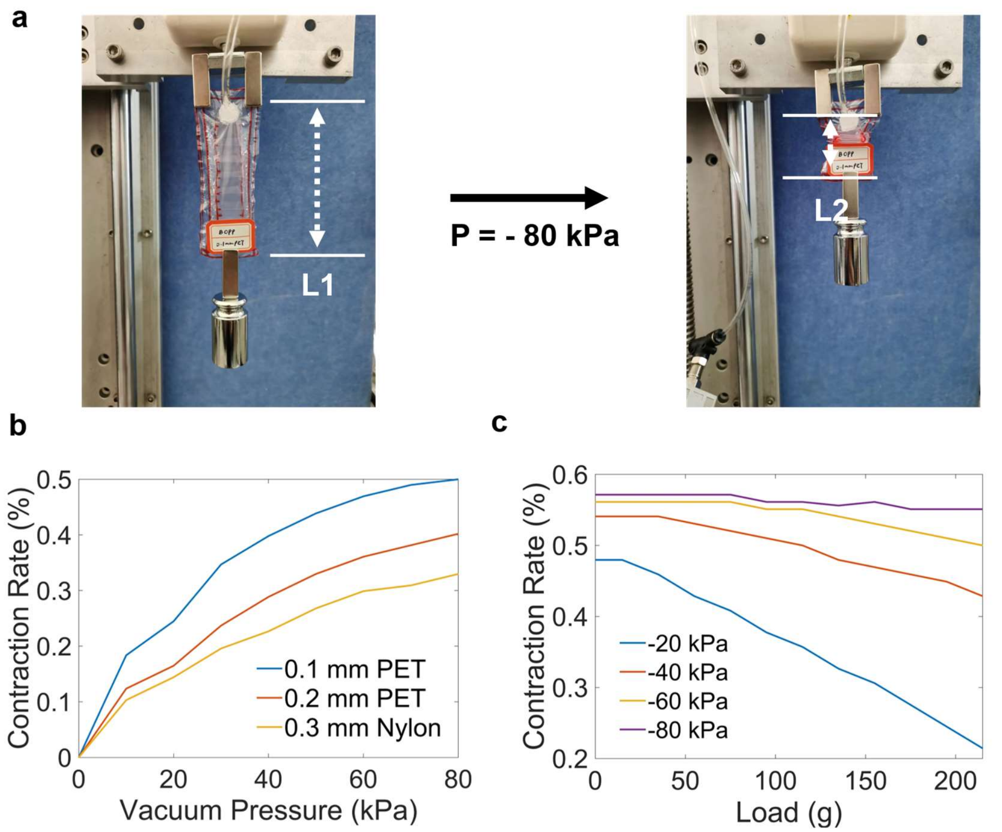


Figure S3. Relationship between load, vacuum pressure, skeleton materials and contraction rate. (a) States before and after a vacuum pressure (-80kPa) is applied to the pneumatic origami actuator. (b) Relationship between vacuum pressure and contraction rate with respect of different skeleton thickness. (c) Relationship between load and contraction rate with respect of different vacuum pressure.

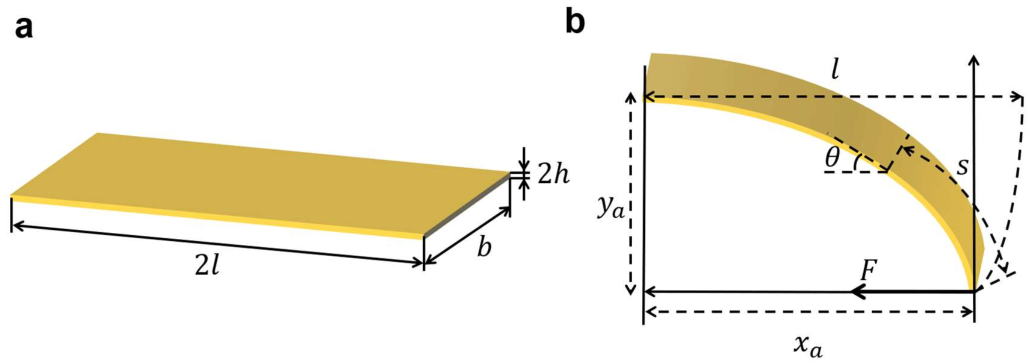


Figure S4. Geometric deformation schematic of the MFC sheet. (a) Characteristics of the MFC piece, including the length $2l$, the width b and the thickness $2h$. (b) Characteristics of the deformed MFC piece, including the deformation on x-axis x_a and y-axis y_a, the angle θ with distance of s from the endpoint.

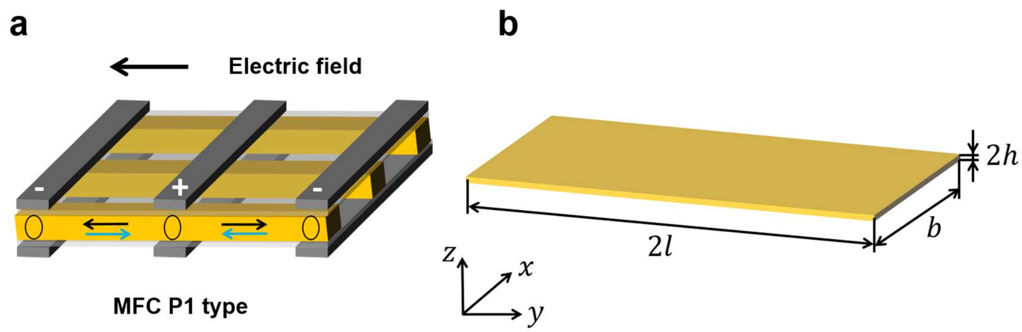


Figure S5. Schematic of the piezo-electric output force. (a) Structure and working principle of the P1-type MFC. (b) Coordinate system on the MFC.

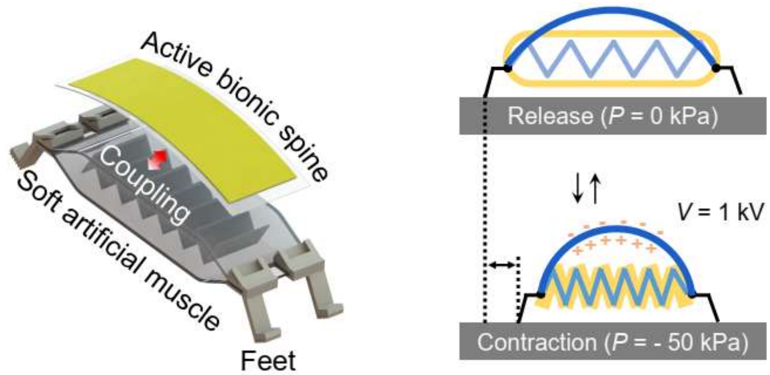


Figure S6. Illustration of the mechanical structure and motion process of the spinal crawling robot.

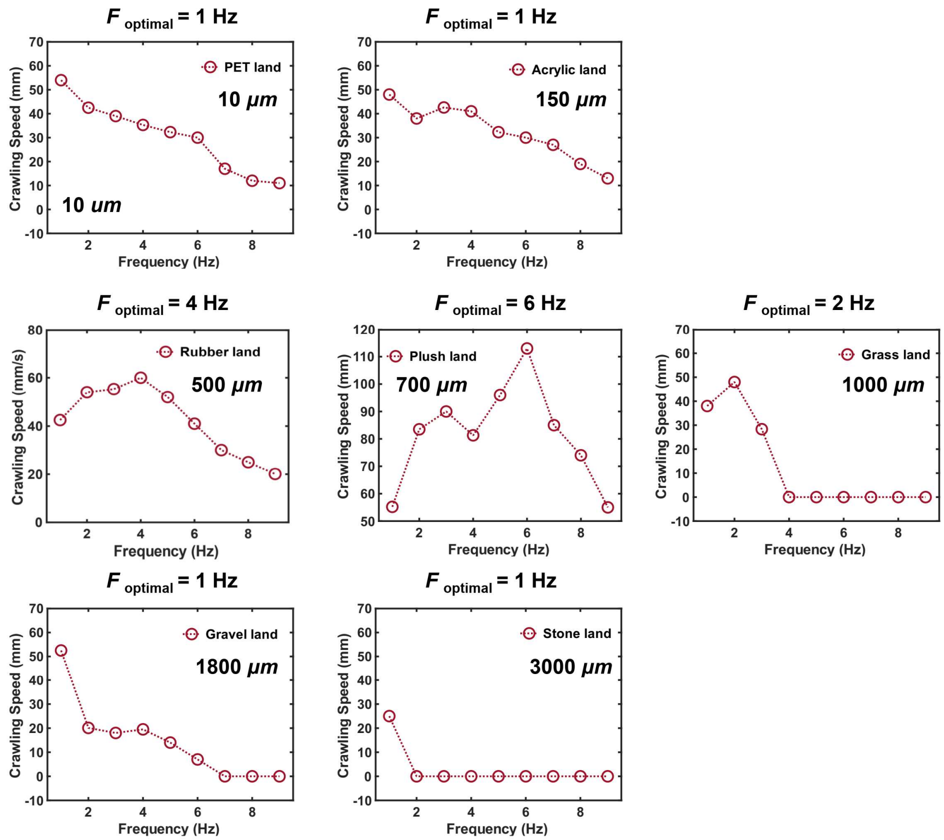


Figure S7. Measured crawling speed (mm/s) of robot as a function of actuation frequency on different terrains.

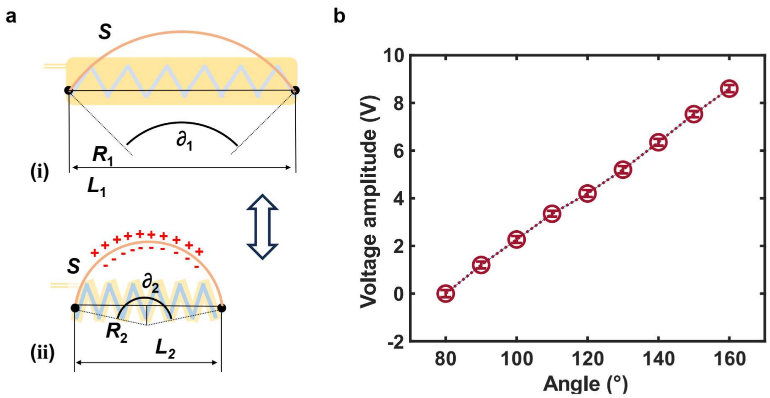


Figure S8. Modeling and characterization of the crawling motion. (a) The geometrical relationship between the spinal bending angle and the robot body length of the crawling robot at different motion states. (b) Measured output voltage amplitude as a function of the bending angle of the active bionic spine.

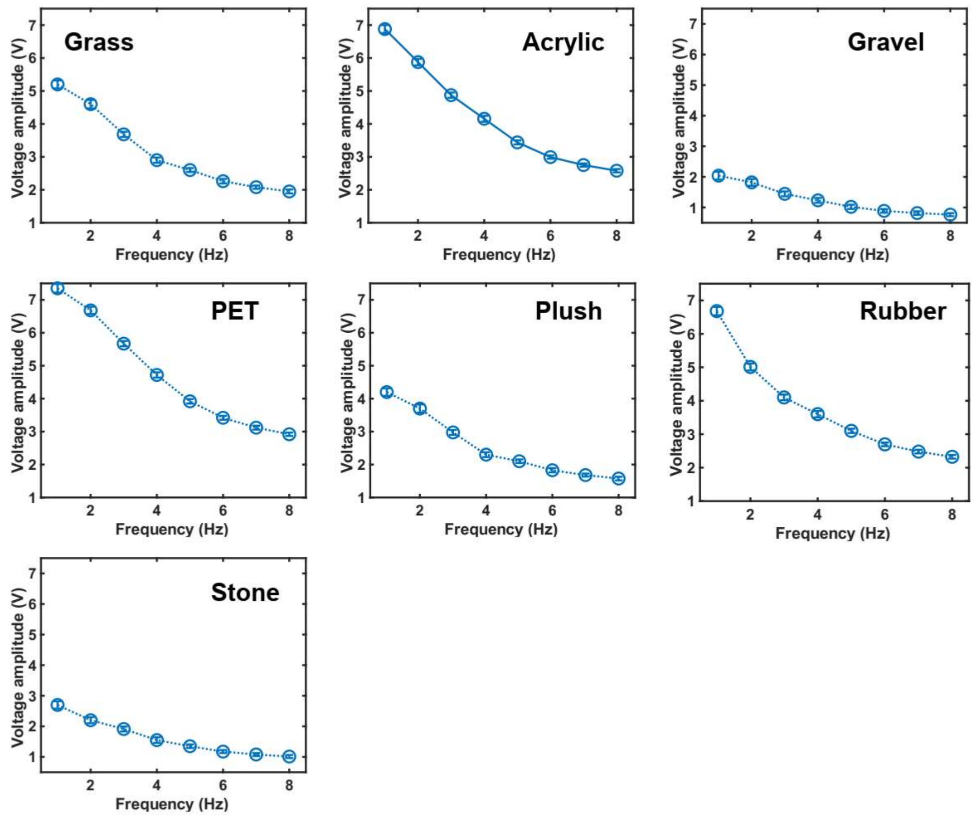


Figure S9. Measured output voltage amplitude of the crawling robot on various terrains with different actuation frequency.

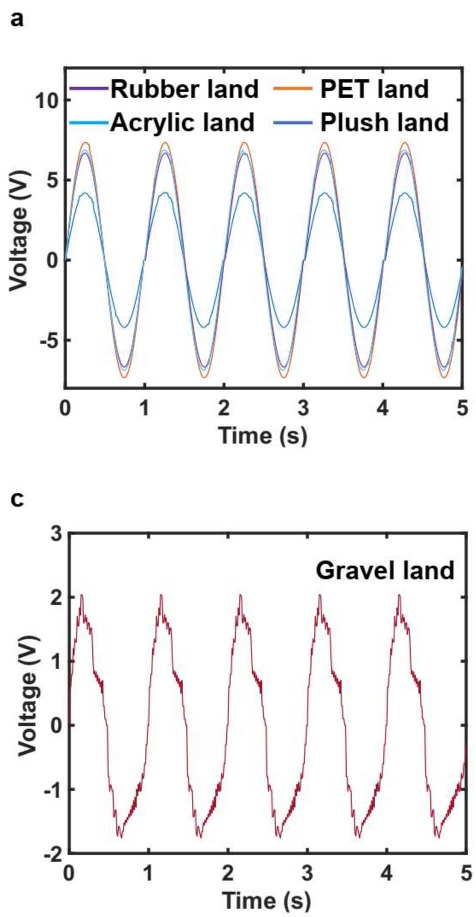


Figure S10. Monitored feedback signal waveform of the bionic spine when the crawling robot moves on different terrains with an actuation frequency of 1 Hz. (a) Four flat lands with different surface roughness, including: Rubber land, PET land, Acrylic land and Plush land. (b) Grass land. (c) Gravel land. (d) Stone land.

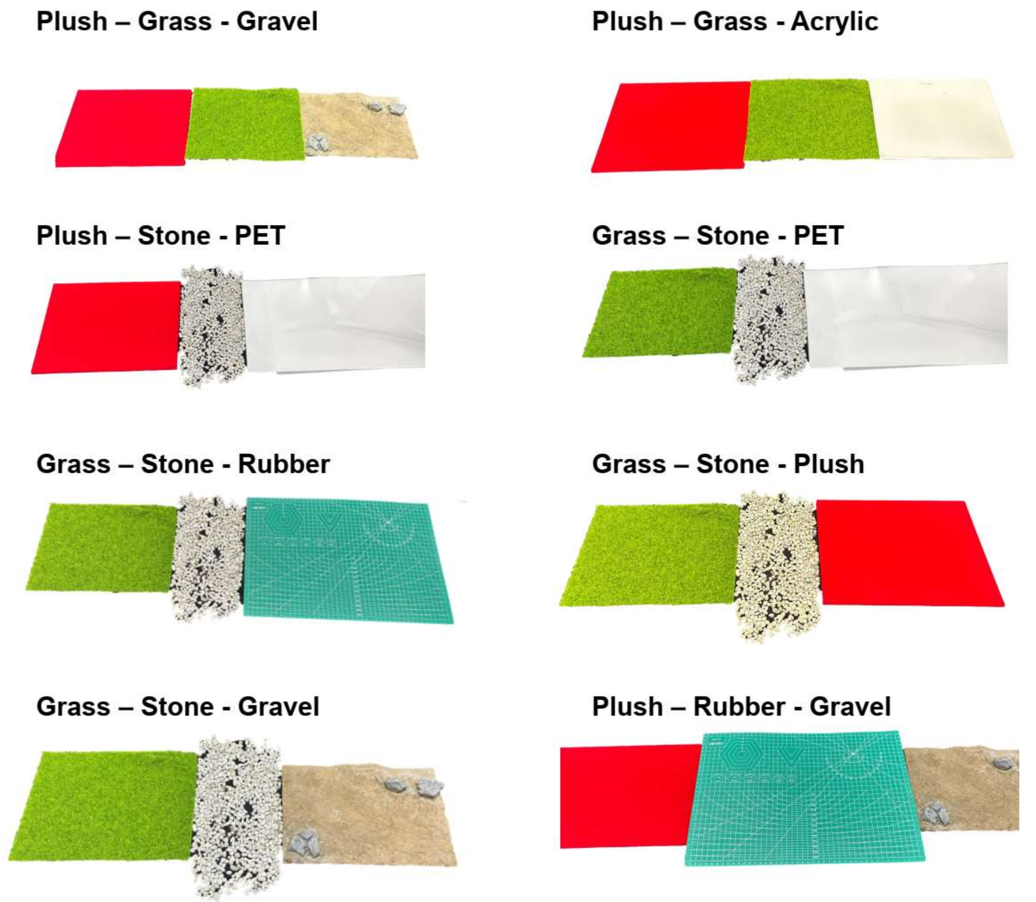


Figure S11. Images of the multi-terrain recognition tests.

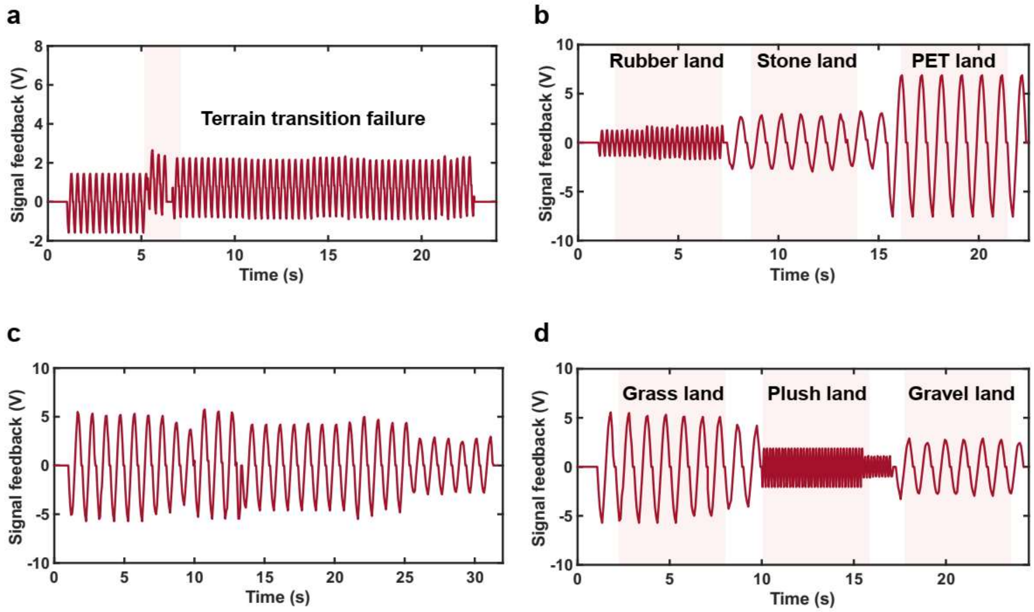


Figure S12. Monitored bionic spine feedback signal during the self-adaptation process of the crawling robot. (a) Feedback signal of the unimodal crawling robot for task 1. (b) Feedback signal of the self-adaptive crawling robot for task 1. (c) Feedback signal of the unimodal crawling robot for task 2. (d) Feedback signal of the self-adaptive crawling robot for task 2.

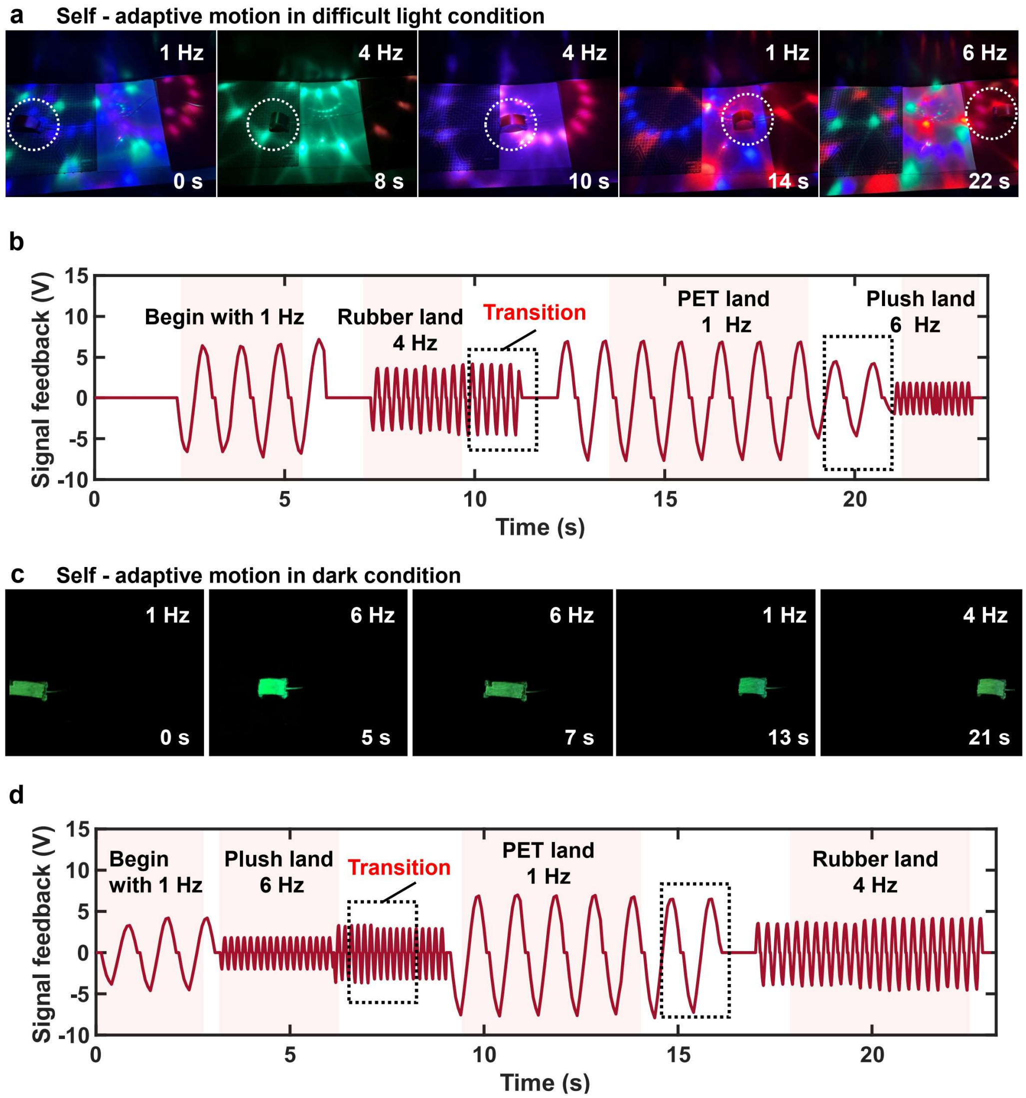


Figure S13. Terrain recognition and self-adaptive motion of the intelligent crawling robot in complex light conditions. (a) Images of the crawling robot moving in a dazzling light condition with self-adaptation. (b) Monitored feedback signal from e-skin and terrain recognition results as a function of time during the whole motion process in a dazzling light condition. (c) Images of the crawling robot moving in a purely dark condition with self-adaptation. (d) Monitored feedback signal from e-skin and terrain recognition results as a function of time during the whole motion process in a purely dark condition.

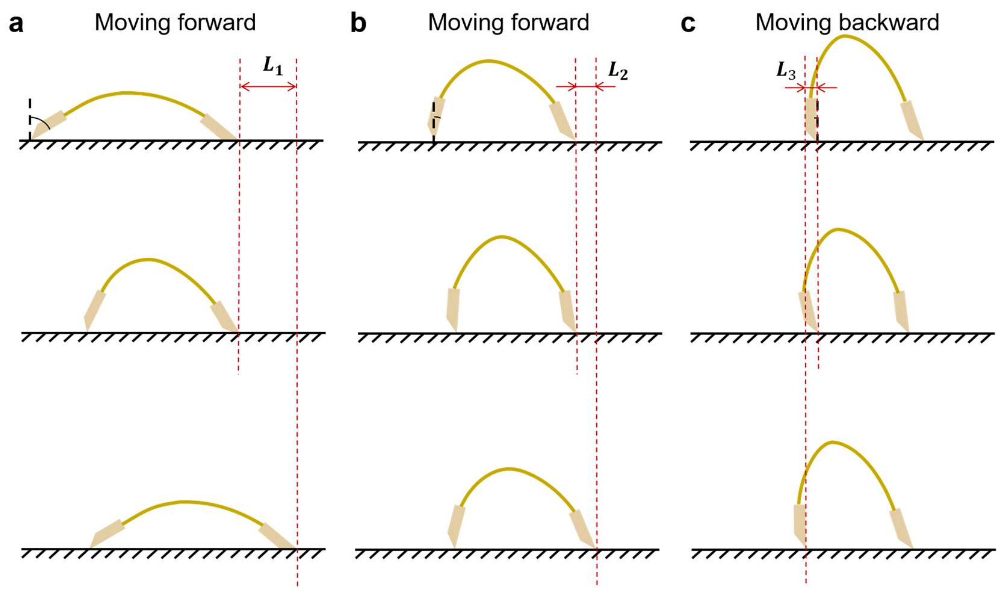


Figure S14. Gait of amphibious robots at different frequencies. (a) Low frequency (1~2 Hz). (b) Middle frequency (2~8 Hz). (c) High frequency (>8 Hz).

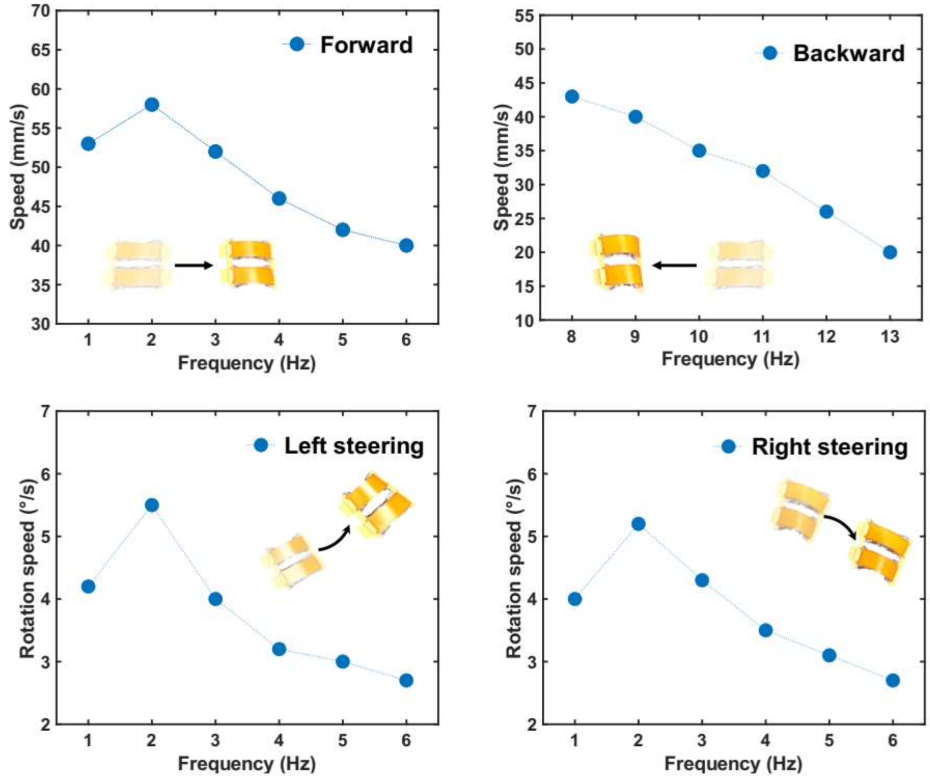


Figure S15. Measured crawling and rotation speed of the omnidirectional robot as a function of actuation frequencies on the rubber land.

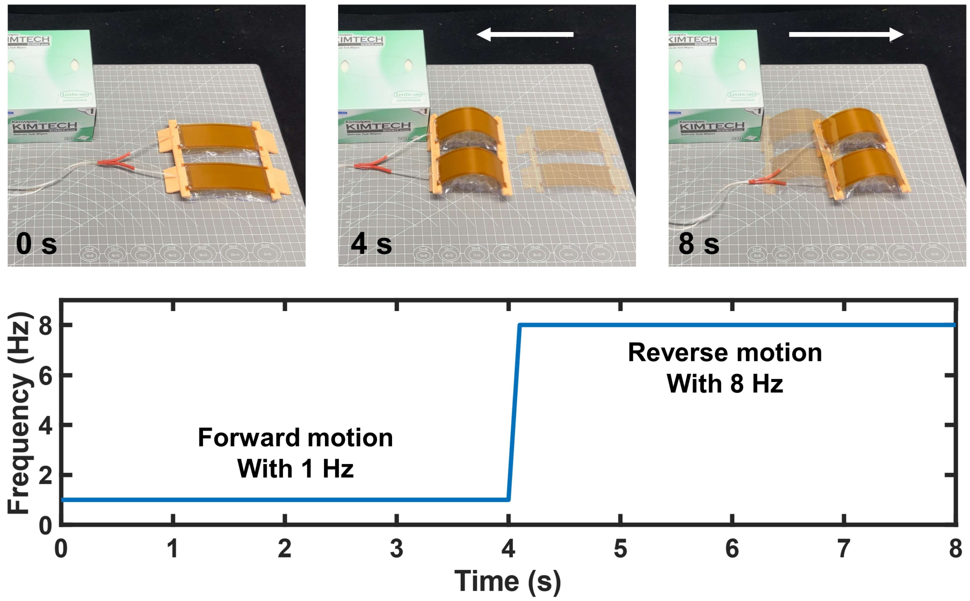


Figure S16. Obstacle avoiding process and actuating frequencies in forward motion process and reverse motion process.

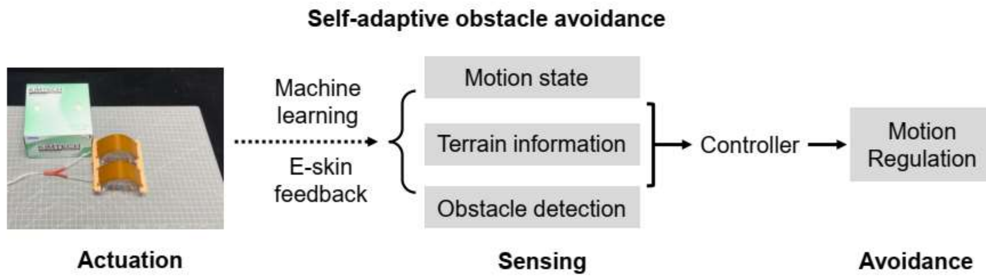


Figure S17. Demonstration of the self-adaptive obstacle process and methodology of the intelligent soft robot.

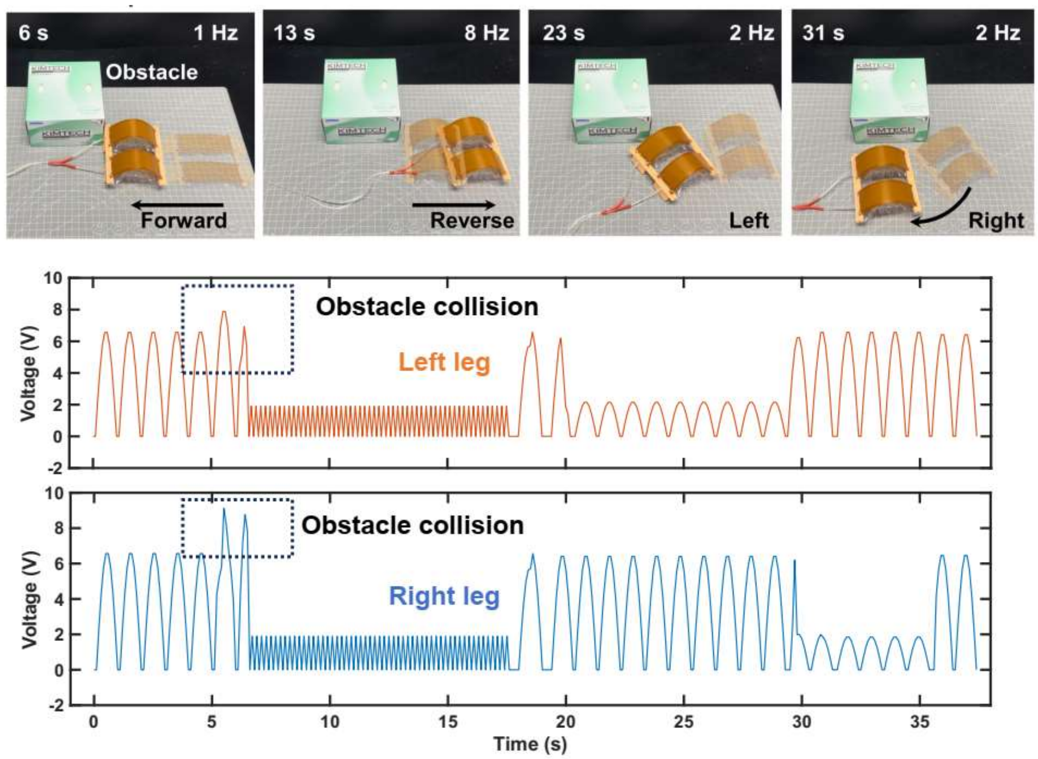


Figure S18. Images of the intelligent robot performing a self-adaptive obstacle avoidance task and the monitored feedback signal from e-skin as a function of time during the whole motion process.

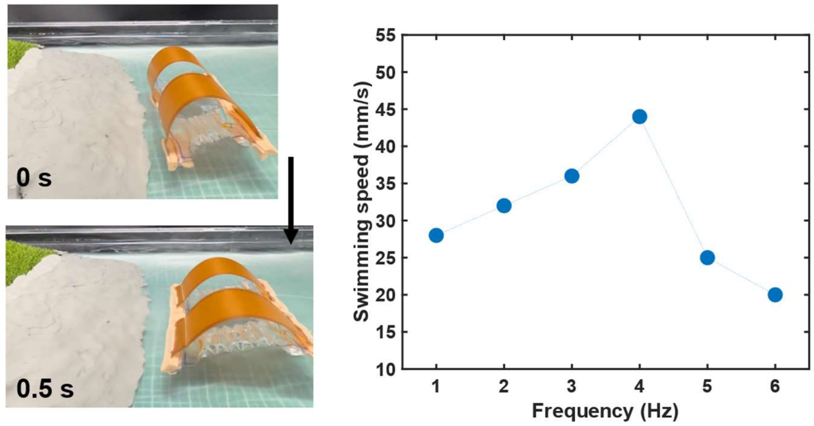


Figure S19. Images of the amphibious robot moving in the aquatic area and the measured swimming speed of the amphibious robot as a function of actuation frequencies.

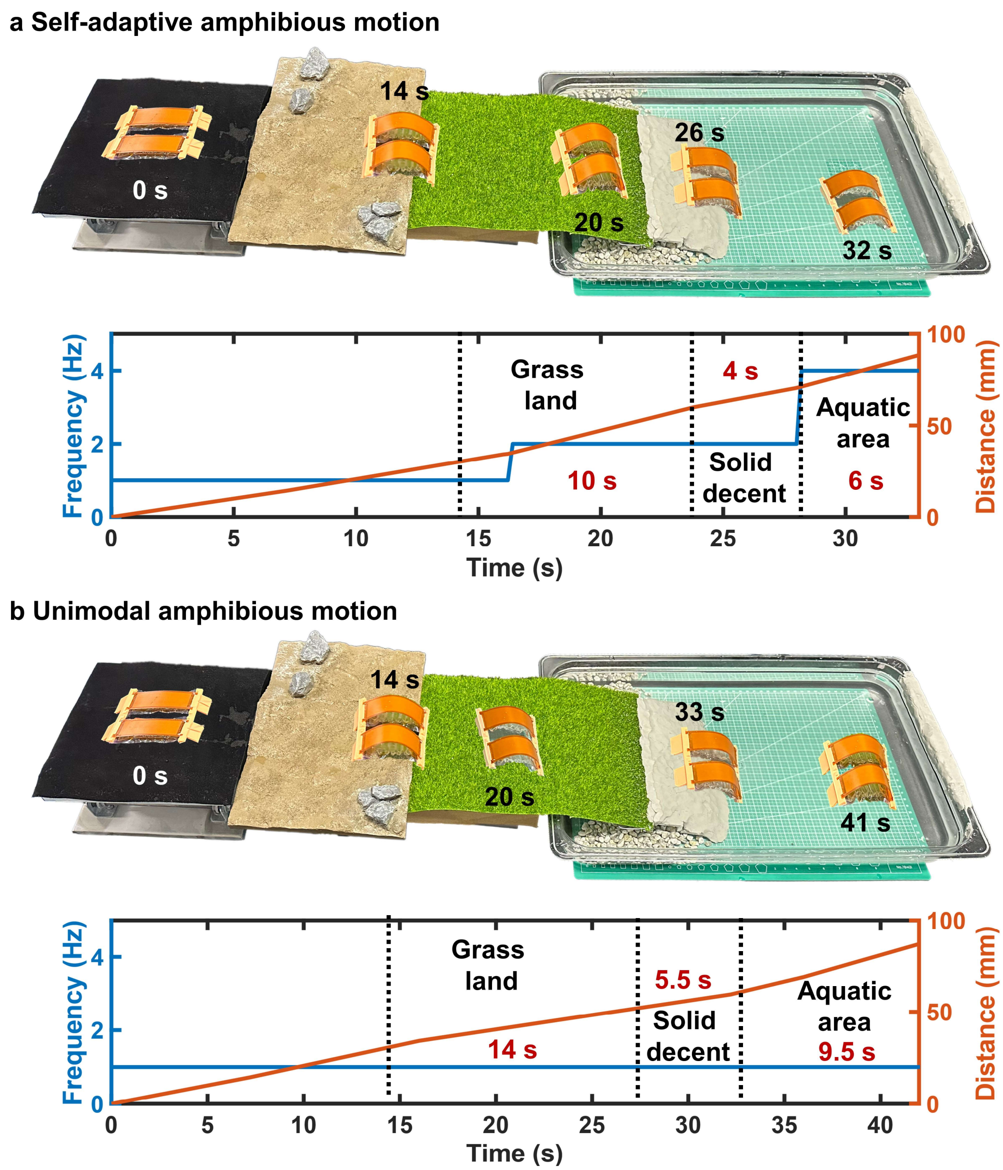


Figure S20. Demonstration of the amphibious soft robot. (a) With self-adaption process. (b) Without self-adaption process.

Legends for Videos S1 to S6:

Supplementary Video 1: Self-adaptability enables high task-successful rate.

Supplementary Video 2: Self-adaptability enables high task efficiency.

Supplementary Video 3: Self-adaption and terrain recognition in dazzling light conditions.

Supplementary Video 4: Self-adaption and terrain recognition in dark conditions.

Supplementary Video 5: Self-adaptive obstacle avoidance.

Supplementary Video 6: Self-adaptive amphibious motion

References

1 Rus, D., and Tolley, M.T. (2015). Design, fabrication and control of soft robots. Nature 521(7553):467-475. https://doi.org/10.1038/nature14543.

2 Xiong, J., Chen, J., and Lee, P.S. (2021). Functional Fibers and Fabrics for Soft Robotics, Wearables, and Human-Robot Interface. Adv. Mater. 33(19):2002640. 2002640, https://doi.org/10.1002/adma.202002640.

3 Han, C., Jeong, Y., Ahn, J., et al. (2023). Recent Advances in Sensor-Actuator Hybrid Soft Systems: Core Advantages, Intelligent Applications, and Future Perspectives. Adv. Sci.:2302775. https://doi.org/10.1002/advs.202302775.

4 Li, G., Chen, X., Zhou, F., et al. (2021). Self-powered soft robot in the Mariana Trench. Nature 591(7848):66-71. https://doi.org/10.1038/s41586-020-03153-z.

5 Li, M., Pal, A., Aghakhani, A., et al. (2022). Soft actuators for real-world applications. Nat. Rev. Mater. 7(3):235-249. https://doi.org/10.1038/s41578-021-00389-7.

6 Kim, S., Laschi, C., and Trimmer, B. (2013). Soft robotics: a bioinspired evolution in robotics. Trends Biotechnol. 31(5):287-294. https://doi.org/10.1016/j.tibtech.2013.03.002.

7 Zhang, Y., Zhang, W., Gao, P., et al. (2022). Finger-palm synergistic soft gripper for dynamic capture via energy harvesting and dissipation. Nat. Commun. 13(1):7700. 7700, https://doi.org/10.1038/s41467-022-35479-9.

8 Ma, K., Chen, X., Zhang, J., et al. (2023). Inspired by Physical Intelligence of an Elephant Trunk: Biomimetic Soft Robot With Pre-Programmable Localized Stiffness. IEEE Robot. Autom. Lett. 8(5):2898-2905. https://doi.org/10.1109/LRA.2023.3256922.

9 Yao, T., Kos, Z., Zhang, Q.X., et al. (2022). Nematic Colloidal Micro-Robots as Physically Intelligent Systems. Adv. Funct. Mater. 32(44):2205546. 2205546, https://doi.org/10.1002/adfm.202205546.

10 Hausladen, M.M., Zhao, B., Kubala, M.S., et al. (2022). Synthetic growth by self-lubricated photopolymerization and extrusion inspired by plants and fungi. Proc. Natl. Acad. Sci. U. S. A. 119(33):e2201776119. https://doi.org/10.1073/pnas.2201776119.

11 Zhao, Y., Chi, Y., Hong, Y., et al. (2022). Twisting for soft intelligent autonomous robot in unstructured environments. Proc. Natl. Acad. Sci. U. S. A. 119(22):e2200265119. e2200265119, https://doi.org/10.1073/pnas.2200265119.

12 Sun, J., Lerner, E., Tighe, B., et al. (2023). Embedded shape morphing for morphologically adaptive robots. Nat. Commun. 14(1):6023. https://doi.org/10.1038/s41467-023-41708-6.

13 Baines, R., Patiballa, S.K., Booth, J., et al. (2022). Multi-environment robotic transitions through adaptive morphogenesis. Nature 610(7931):283-289. https://doi.org/10.1038/s41586-022-05188-w.

14 Ilami, M., Bagheri, H., Ahmed, R., et al. (2021). Materials, Actuators, and Sensors for Soft Bioinspired Robots. Adv. Mater. 33(19):2003139.

https://doi.org/10.1002/adma.202003139.
15 Gu, G., Zhang, N., Xu, H., et al. (2023). A soft neuroprosthetic hand providing simultaneous myoelectric control and tactile feedback. Nat. Biomed. Eng 7(4):589-598. https://doi.org/10.1038/s41551-021-00767-0.
16 Dai, J., Li, X.-G., Zhang, T.-Y., et al. (2024). Illuminating a bacterial adaptation mechanism: Infrared-driven cell division in deep-sea hydrothermal vent environments. The Innovation Geoscience 2(1):100050. https://doi.org/10.59717/j.xinn-geo.2024.100050.
17 Tan, D., and Xu, B. (2023). Advanced Interfacial Design for Electronic Skins with Customizable Functionalities and Wearability. Adv. Funct. Mater.:2306793. https://doi.org/10.1002/adfm.202306793.
18 Peng, X., Dong, K., Ye, C., et al. (2020). A breathable, biodegradable, antibacterial, and self-powered electronic skin based on all-nanofiber triboelectric nanogenerators. Sci. Adv. 6(26):eaba9624. eaba9624, https://doi.org/10.1126/sciadv.aba9624.
19 Zou, Z., Zhu, C., Li, Y., et al. (2018). Rehealable, fully recyclable, and malleable electronic skin enabled by dynamic covalent thermoset nanocomposite. Sci. Adv. 4(2):eaaq0508. eaaq0508, https://doi.org/10.1126/sciadv.aaq0508.
20 Park, J., Kim, D., and Kim, Y.T. (2021). Soft and transparent triboelectric nanogenerator based E-skin for wearable energy harvesting and pressure sensing. Nanotechnology 32(38):385403. 385403, https://doi.org/10.1088/1361-6528/ac0c3f.
21 Guo, N., Han, X., Liu, X., et al. (2023). Autoencoding a Soft Touch to Learn Grasping from On-Land to Underwater. Adv. Intell. Syst. 5(3):2200269. https://doi.org/10.1002/aisy.202300382.
22 Wang, X., Fang, G., Wang, K., et al. (2020). Eye-in-Hand Visual Servoing Enhanced With Sparse Strain Measurement for Soft Continuum Robots. IEEE Robot. Autom. Lett. 5(2):2161-2168. https://doi.org/10.1109/LRA.2020.2969953.
23 Liu, X., Onal, C.D., and Fu, J. (2023). Reinforcement Learning of CPG-Regulated Locomotion Controller for a Soft Snake Robot. IEEE Trans. Robot. 39(5):3382-3401. https://doi.org/10.1109/TRO.2023.3286046.
24 Hawkes, E.W., Blumenschein, L.H., Greer, J.D., and Okamura, A.M. (2017). A soft robot that navigates its environment through growth. Sci. Robot. 2(8):eaan3028. https://doi.org/10.1126/scirobotics.aan3028.
25 Huang, C., Xu, T., Liu, J., et al. (2019). Visual Servoing of Miniature Magnetic Film Swimming Robots for 3-D Arbitrary Path Following. IEEE Robot. Autom. Lett. 4(4):4185-4191. https://doi.org/10.1109/LRA.2019.2931234.
26 Xu, F., Wang, H., Liu, Z., et al. (2021). Visual Servoing Pushing Control of the Soft Robot with Active Pushing Force Regulation. Soft Robot. 9(4):690-704. https://doi.org/10.1089/soro.2020.0178.
27 Yang, X., Lan, L., Pan, X., et al. (2023). Bioinspired soft robots based on

organic polymer-crystal hybrid materials with response to temperature and humidity. Nat. Commun. 14(1):2287. 2287, https://doi.org/10.1038/s41467-023-37964-1.

28 Wang, C., Wu, Y., Dong, X., et al. In situ sensing physiological properties of biological tissues using wireless miniature soft robots. Sci. Adv. 9(23):eadg3988. https://doi.org/10.1126/sciadv.adg3988.

29 Su, Q., Zou, Q., Li, Y., et al. A stretchable and strain-unperturbed pressure sensor for motion interference–free tactile monitoring on skins. Sci. Adv. 7(48):eabi4563. https://doi.org/10.1126/sciadv.abi4563.

30 Lin, W., Zhang, D., Lee, W.W., et al. Super-resolution wearable electrotactile rendering system. Sci. Adv. 8(36):eabp8738. https://doi.org/10.1126/sciadv.abp8738.

31 Li, D., Zhou, J., Yao, K., et al. Touch IoT enabled by wireless self-sensing and haptic-reproducing electronic skin. Sci. Adv. 8(51):eade2450. https://doi.org/10.1126/sciadv.ade2450.

32 Liu, W., Duo, Y., Liu, J., et al. (2022). Touchless interactive teaching of soft robots through flexible bimodal sensory interfaces. Nat. Commun. 13(1):5030. https://doi.org/10.1038/s41467-022-32702-5.

33 Jiang, S., Zhang, T., Zhou, Y., et al. (2023). Wearable ultrasound bioelectronics for healthcare monitoring. The Innovation 4(4). https://doi.org/10.1016/j.xinn.2023.100447.

34 Zhang, T., Liu, N., Xu, J., et al. (2023). Flexible electronics for cardiovascular healthcare monitoring. The Innovation 4(5):100485. DOI: https://doi.org/10.1016/j.xinn.2023.100485.

35 Shahsavan, H., Aghakhani, A., Zeng, H., et al. (2020). Bioinspired underwater locomotion of light-driven liquid crystal gels. Proc. Natl. Acad. Sci. U. S. A. 117(10):5125-5133. https://doi.org/10.1073/pnas.1917952117.

36 Truby, R.L., Santina, C.D., and Rus, D. (2020). Distributed Proprioception of 3D Configuration in Soft, Sensorized Robots via Deep Learning. IEEE Robot. Autom. Lett. 5(2):3299-3306. https://doi.org/10.1109/LRA.2020.2976320.

37 Jin, L., Li, Z., Liu, Z., et al. (2022). Flexible unimodal strain sensors for human motion detection and differentiation. npj Flex. Electron. 6(1):74. https://doi.org/10.1038/s41528-022-00205-4.

38 Kim, M., Yu, A., Kim, D., et al. (2023). Multi-Agent Control of Laser-Guided Shape-Memory Alloy Microrobots. Adv. Funct. Mater.:2304937. https://doi.org/10.1002/adfm.202304937.

39 Hu, D., Giorgio-Serchi, F., Zhang, S., and Yang, Y. (2023). Stretchable e-skin and transformer enable high-resolution morphological reconstruction for soft robots. Nat. Mach. Intell. 5(3):261-272. https://doi.org/10.1038/s42256-023-00622-8.

40 Shih, B., Shah, D., Li, J., et al. (2020). Electronic skins and machine learning for intelligent soft robots. Sci. Robot. 5(41):eaaz9239. https://doi.org/10.1126/scirobotics.aaz9239.

41 Liu, W., Duo, Y., Chen, X., et al. (2023). An Intelligent Robotic System

Capable of Sensing and Describing Objects Based on Bimodal, Self-Powered Flexible Sensors. Adv. Funct. Mater. 33(41):2306368. https://doi.org/10.1002/adfm.202306368.
42 Deng, M., Fan, F., and Wei, X. (2023). Learning-Based Object Recognition via a Eutectogel Electronic Skin Enabled Soft Robotic Gripper. IEEE Robot. Autom. Lett. 8(11):7424-7431. https://doi.org/10.1109/LRA.2023.3316096.
43 Thuruthel, T.G., Shih, B., Laschi, C., and Tolley, M.T. (2019). Soft robot perception using embedded soft sensors and recurrent neural networks. Sci. Robot. 4(26):eaav1488. https://doi.org/10.1126/scirobotics.aav1488.
44 Jin, T., Sun, Z., Li, L., et al. (2020). Triboelectric nanogenerator sensors for soft robotics aiming at digital twin applications. Nat. Commun. 11(1):5381. https://doi.org/10.1038/s41467-020-19059-3.
45 Xie, M., Zhu, M., Yang, Z., et al. (2021). Flexible self-powered multifunctional sensor for stiffness-tunable soft robotic gripper by multimaterial 3D printing. Nano Energy 79:105438. https://doi.org/10.1016/j.nanoen.2020.105438.
46 Gong, S., Ding, Q., Wu, J., et al. (2022). Bioinspired Multifunctional Mechanoreception of Soft–Rigid Hybrid Actuator Fingers. Adv. Intell. Syst. 4(5):2100242. https://doi.org/10.1002/aisy.202100242.
47 Yan, H., Wang, Y., Shen, W., et al. (2022). Cable-Driven Continuum Robot Perception Using Skin-Like Hydrogel Sensors. Adv. Funct. Mater. 32(34):2203241. https://doi.org/10.1002/adfm.202203241.
48 Jiao, Z., Hu, Z., Shi, Y., et al. (2024). Reprogrammable, intelligent soft origami LEGO coupling actuation, computation, and sensing. The Innovation 5(1):100549. https://doi.org/10.1016/j.xinn.2023.100549.
49 Dosch, J. J., Inman, D. J. & Garcia, E. A Self-Sensing Piezoelectric Actuator for Collocated Control. Journal of Intelligent Material Systems and Structures 3, 166-185 (1992). https://doi.org:10.1177/1045389X9200300109
50 Zheng, L. & Weijie, D. Position and force self-sensing piezoelectric valve with hysteresis compensation. Journal of Intelligent Material Systems and Structures 33, 170-182 (2021). https://doi.org:10.1177/1045389X211011677
51 McPherson, T. & Ueda, J. A Force and Displacement Self-Sensing Piezoelectric MRI-Compatible Tweezer End Effector With an On-Site Calibration Procedure. IEEE/ASME Transactions on Mechatronics 19, 755-764 (2014). https://doi.org:10.1109/TMECH.2013.2257827
52 Qian, C. et al. A piezoelectric hydraulic linear motor with velocity self-monitoring. Sens. Actuator A-Phys. 306, 111962 (2020).
53 Li, J. et al. Developments and Challenges of Miniature Piezoelectric Robots: A Review. Adv. Sci. (2023). https://doi.org:10.1002/advs.202305128
54 Ruth, S. R. A., Feig, V. R., Tran, H. & Bao, Z. Microengineering Pressure Sensor Active Layers for Improved Performance. Adv. Funct. Mater. 30, 2003491 (2020). https://doi.org:10.1002/adfm.202003491
55 Wu, Z. et al. A Humidity-Resistant, Sensitive, and Stretchable Hydrogel-Based Oxygen Sensor for Wireless Health and Environmental Monitoring. Adv.

Funct. Mater. 34, 2308280 (2024). https://doi.org:10.1002/adfm.202308280
56 Wang, H. et al. High-Performance Hydrogel Sensors Enabled Multimodal and Accurate Human–Machine Interaction System for Active Rehabilitation. Adv. Mater. n/a, 2309868 (2023). https://doi.org:10.1002/adma.202309868
57 Zhai, K. et al. High-Performance Strain Sensors Based on Organohydrogel Microsphere Film for Wearable Human–Computer Interfacing. Adv. Sci. 10, 2205632 (2023). https://doi.org:10.1002/advs.202205632
58 Liu, W. et al. An Intelligent Robotic System Capable of Sensing and Describing Objects Based on Bimodal, Self-Powered Flexible Sensors. Adv. Funct. Mater. 33 (2023). https://doi.org:10.1002/adfm.202306368
59 Dai, H. et al. Split-Type Magnetic Soft Tactile Sensor with 3D Force Decoupling. Adv. Mater. (2023). https://doi.org:10.1002/adma.202310145
60 Kim, M. et al. Strain-Sensitive Flexible Magnetoelectric Ceramic Nanocomposites. Adv. Mater. Technol. 8 (2023). https://doi.org:10.1002/admt.202202097
61 Gomez, P., Litvinov, D. & Khizroev, S. A method to design high SNR nanoscale magnetic sensors using an array of tunnelling magneto-resistance (TMR) devices. Journal of Physics D: Applied Physics 40, 4396 (2007). https://doi.org:10.1088/0022-3727/40/15/002
62 Li, S., Vogt, D. M., Rus, D. & Wood, R. J. Fluid-driven origami-inspired artificial muscles. Proc. Natl. Acad. Sci. U. S. A. 114, 13132-13137 (2017). https://doi.org:10.1073/pnas.1713450114
63 Mosadegh, B. et al. Pneumatic Networks for Soft Robotics that Actuate Rapidly. Adv. Funct. Mater. 24, 2163-2170 (2014). https://doi.org:10.1002/adfm.201303288
64 Kurumaya, S., Nabae, H., Endo, G. & Suzumori, K. Design of thin McKibben muscle and multifilament structure. Sensors and Actuators A: Physical 261, 66-74 (2017). https://doi.org:10.1016/j.sna.2017.04.047
65 Ching-Ping, C. & Hannaford, B. Measurement and modeling of McKibben pneumatic artificial muscles. IEEE Transactions on Robotics and Automation 12, 90-102 (1996). https://doi.org:10.1109/70.481753
66 Kim, M.-S. et al. Shape Memory Alloy (SMA) Actuators: The Role of Material, Form, and Scaling Effects. Adv. Mater. 35 (2023). https://doi.org:10.1002/adma.202208517
67 Kellaris, N., Gopaluni Venkata, V., Smith, G. M., Mitchell, S. K. & Keplinger, C. Peano-HASEL actuators: Muscle-mimetic, electrohydraulic transducers that linearly contract on activation. Sci. Robot. 3, eaar3276 (2018). https://doi.org:10.1126/scirobotics.aar3276
68 Acome, E. et al. Hydraulically amplified self-healing electrostatic actuators with muscle-like performance. Science 359, 61-65 (2018). https://doi.org:10.1126/science.aao6139
69 Rothemund, P., Kellaris, N., Mitchell, S. K., Acome, E. & Keplinger, C. HASEL Artificial Muscles for a New Generation of Lifelike Robots—Recent Progress and Future Opportunities. Adv. Mater. 33, 2003375 (2021).

https://doi.org:10.1002/adma.202003375
70 Gu, G. Y., Zhu, J., Zhu, L. M. & Zhu, X. Y. A survey on dielectric elastomer actuators for soft robots. Bioinspir. Biomim. 12 (2017). https://doi.org:10.1088/1748-3190/12/1/011003